The

DREAMLESS MACHINE



GAUSS

THE

DREAMLESS MACHINE

Gauss

Revised Edition

Preface

I often think of an evening when there is still no answer. The lamp is on; words and figures have been written down and crossed out; the people outside have gone home. Someone sits at a desk, knowing the calculation may be wrong, yet unwilling to put those few pages away. In writing this novel, I wanted to come close to such evenings, and to see how, the next morning, that person would go on living.

More than a century separates Einstein and Lin Che. The old records Noah preserves bring their lives together. When writing about Einstein, I wanted to set aside the conclusions we already know and return to a time when he still worried about finding a position and lost his temper over an objection. He did not yet know that his name would appear in textbooks. Nor did the people sitting across from him. Planck was willing to read those few manuscript pages, first of all, because the question on them was worth reading.

By Lin Che's time, a fluent paper or a plausible explanation can be produced very quickly. He has to work harder to establish what the results in his hands actually show. He also needs a salary, a contract, and enough of someone else's time for them to read his manuscript to the end. When he gains the power to distribute those things himself, he may do no better than the people who came before him. I did not want one elegant discovery to let him escape these difficulties. A researcher still has to return to other people and answer for what he has said.

Mileva and Fang Xiuqin spend many such days beside them. One can carry on the discussion of a problem on the page; the other cannot even understand its terms. Their love takes different forms, and each has her own weariness and hurt. I wanted to write more of their lives, too: the little booklet Mileva has not finished, the time Fang Xiuqin finally learns how to turn her head and breathe while swimming. These things also deserve to be heard to the end.

The novel draws on Einstein's life and certain debates in the history of science. Its private conversations and domestic details include literary invention; Noah's involvement is itself a fictional premise. Lin Che's complex-spacetime proposal is likewise an imagined theory. He hopes to account, through a single set of laws, for the deterministic evolution described by general relativity and the probabilistic outcomes of quantum field theory. Many steps along that road remain to be proved and tested. The hopes expressed in the story are not scientific conclusions.

I also want to know why, when a machine can carry on calculating, a person still comes back to the desk. Sometimes Lin Che no longer believes his own explanations. Yet after putting away a page of mistaken calculations, he finds himself thinking about it on the way home. Those distractions, frustrations, and occasional dreams belong to this book as well.

The story continues on the back of a sheet of paper. Twelve-year-old Albert copies down the question Newton left behind, turns the page over, and remembers his compass. He does not yet know how he will answer it. He remembers only that the needle will never stay where he turns it.

Gauss

Contents

Preface i

Part One Einstein 1

1 The Compass 1

2 Chasing Light 4

3 Borrowed Notes 6

4 Mileva's Notebook 8

5 The Patent Office Lamp 12

6 The Olympia Academy 15

7 1905 19

8 A Letter from Berlin 22

9 General Relativity 25

10 After Fame 29

11 The Last Equation 33

12 The Unfinished Page 36

Contents

Part Two Lin Che 40

13 A Voice in the Store Room 40

14 Complex Spacetime 43

15 The Successor 48

16 Later 51

17 The Missing Names 56

18 A Room 59

19 Output 63

20 Another Kind of Excellence 67

21 Page Four 71

22 After School 74

23 Those Who Left 77

24 Unfinished 84

Afterword One Those Who Dream 89

Afterword Two Noah's Monologue 96

Part One Einstein

Chapter One The Compass

In 1891, twelve-year-old Albert stood beside a workbench in his father's factory, holding a sheet of paper written in German.

The box on the bench sat between several coils and a broken meter. It was scarcely larger than a thick dictionary. His father had checked the connections; his uncle had secured a loose wire. Patches of different colours covered the casing, and one had a sliver of wood wedged beneath it. The box had arrived with a consignment of old electrical equipment, but the papers accompanying it did not clearly identify its previous owner.

His uncle tapped the inspection panel. A very faint voice came from inside. He bent closer, and the voice repeated itself: please bring the paper up to the narrow slot.

Albert offered it an arithmetic problem. The answer came quickly, followed by a sentence that had nothing to do with the problem.

“What is the origin of gravity?”

His father thought the wiring must still be faulty and told it to repeat the words. It did, then explained that this was an old record it could still read.

“Who asked that?” Albert said.

“Isaac Newton.”

His uncle laughed and asked when Newton had visited this factory. The box did not pursue the joke. It explained only that the sentence had been transferred many times, and the surviving records were insufficient to reconstruct the conversation. It gave a year long before his father's grandfather had been born. His uncle picked up the meter and took another reading.

Someone outside was asking when a delivery would be ready. His father told Albert not to touch anything inside the box. His uncle put the tools away, and the two men left. Albert promised he would not, and copied the question beneath his arithmetic.

He knew Newton's name. He had also heard explanations of why things fell to the ground. But the box's question seemed to begin again at the point where the explanation had ended.

Outside, wheels rattled over the stone street. A draught lifted the corner of the paper. He held it down, then turned it over. The back was blank. He sat for a while, thinking of the compass at home.

That evening, his father let him take the box home, telling him to keep it on the table and disconnect the external power supply when he was not using it at night. Albert carried it in both arms and walked more slowly than usual. In his room, he took out his old compass and set it beside the box. The needle shifted. He moved the compass away, then brought it back. The needle moved with it.

“You push it too,” he said.

“My operation produces electromagnetic effects.”

“Are you touching it?”

“No.”

“Then what's between you?”

The box explained magnetic fields. He listened for a while, then moved the compass to the far side of the table.

“You call it a magnetic field. But I still don't know how it makes the needle move from so far away.”

One face of the box lit up, displaying a diagram with arrows. He copied a few strokes, then drew two dots on the back of the paper. If there was nothing you could touch between them, he asked, how did one dot know the other had moved?

“You must first explain what you mean by ‘know.’”

“You explain what you mean by ‘between’ first.”

The box explained that gravity and magnetism could not be treated as the same interaction. He listened, writing the two explanations separately, still moving his finger back and forth between the dots. Only when his mother came in did he realise he had missed dinner. She felt the box to see whether it was hot and told him to come downstairs.

A few days later, his father found an old copy of half an assembly drawing beneath the packing material. Its wooden frame was unlike the present metal casing, though one of the contacts occupied much the same position. Albert showed it to the box. He made out the name Leonardo and a note about transferring the core.

The name also appeared in the old material the box could identify. Those instructions concerned supports, protective panels, and contacts. There was still no reliable answer to the question of who had made the thing at the centre. The drawing would not enable his father to build a second one. Nor did the paper explain how it remembered all those words.

“Are you from the future?” Albert asked.

“I cannot verify that.”

“Then why did the people in the future make you so useless?”

“At present I cannot pass on your complaint.”

Albert laughed aloud.

He and his uncle tried lifting the outermost inspection panel. Beneath it were old rivets, repaired supports, and wiring added later. They could draw everything they were able to remove, but the dark grey core at the centre had no seam they recognised. His father allowed them to replace a loose external terminal. He would not let them prise any further.

“If we could build a second one…” his father said.

The next day he went back to repairing a generator that had to be delivered on time. Albert saw a new entry on the inventory: provenance unknown; held pending further investigation. His treasure was still listed alongside old parts awaiting disposal. He slipped the half-drawing into his notebook. Before it was Newton's question; on the reverse were the two dots he had drawn.

When his mother came in later, papers covered the table. First she asked whether the box might catch fire. Then she asked whether it could remind her son to go to bed.

It said it could.

At nine that evening, as agreed, the box told him it was time to sleep. He said in a minute. It continued to remind him every few minutes. When he told it to stop, it asked whether his mother's earlier instruction had been withdrawn.

He named it Noah. The box contained old knowledge, unfamiliar symbols, and memories of uncertain origin, like a boat that had drifted in from somewhere no one knew. After a few days, he no longer called it “the box.”

Noah drew up a study plan for him. It was detailed and sensibly ordered, and would, it said, prevent a great deal of duplicated effort. Albert followed it for three days. On the fourth, he picked up the compass again.

“We have finished the lesson on magnetic fields,” Noah reminded him.

“I haven't understood it yet.”

“You can already answer the relevant questions.”

The boy turned the needle south, let go, and watched it refuse to remain there.

“Not the one I'm asking,” he said.

Chapter Two Chasing Light

At sixteen, Albert failed the entrance examination.

Noah had long since noticed that he was spending far too little time revising languages and natural history. It had reminded him and prepared exercises. Albert always said in a minute, then went back to physics. After the examination, on the train, he folded his results and tucked them behind the last page of a book, where they would not show.

“You can say ‘I told you so’ now.”

“That would not change your marks.”

Albert patted it through the cloth bag by way of thanks.

He went to Aarau to continue his schooling. At the house where he boarded, people could discuss questions over dinner. They could also disagree. At first his objections came out too forcefully, as though he expected to be punished every time. When he discovered that no one made him stand up for it, his voice gradually relaxed.

His room faced the street. Early in the morning, carts passed and their wheels made the windowpanes rattle. At night, when he lit the lamp, his own reflection appeared in the glass. He had left his family, rejoined them, and left again. His clothes hung in the wardrobe, yet sometimes, waking in the night, he still reached towards where the door had been in his old room.

Noah stood beside the bed. When he called, it lit up.

“What do you need?” the machine asked.

“Nothing.”

“Then I shall return to silent mode.”

He opened his mouth, then pulled the blanket a little higher.

Noah once corrected a French composition for him. The sentences flowed so smoothly that even he found them unfamiliar. When the teacher asked why he had used a particular word, he could not answer. Back in his room, he rewrote the composition with simpler words and several awkward passages. The next day the teacher asked about the same passage. When Albert finished explaining, he found he had one more thing to say.

That year he often walked by the river. One evening, near dusk, he stopped and asked Noah what he would see if he chased a beam of light at the speed of light.

Noah first described a light wave apparently frozen before him, then pointed out the conflict between that description and electromagnetic theory. Albert crouched down and made a few marks on the ground. A carriage passed, its wheel erasing one of the lines.

“If two carriages travel at the same speed, one looks stationary from the other,” he said.

“Light may not permit the same treatment.”

“Why not?”

Noah showed him several mathematical relations. He understood only some of them. Of the accounts of motion the machine found, some lacked sources; others rested on different assumptions. He asked which he could use. None disappeared from the screen.

His legs had gone numb. He stood and asked:

“Aren't you supposed to know a great deal?”

“These accounts cannot all be true at once.”

“Choose the most beautiful.”

A compact expression appeared on the screen.

“Why that one?” Albert asked.

“You asked me to select by form.”

He closed the cloth bag and said nothing all the way back. At his lodgings, he took out a sheet of paper and copied the beautiful expression. Beside it he wrote: What does this require me to give up?

The next day he returned to the river. A patch of light on the water was broken by the shadow of a tree. He ran a few steps along the bank. His shoe sank in the mud. He looked down, pulled it free, and when he looked up again, the bright patch had moved.

On the way home, Noah asked whether he had made a new observation.

He said his shoes were wet.

The machine suggested leaving them somewhere well ventilated. He turned them upside down on the windowsill. His socks were soaked too, so he sat barefoot at the desk and went on drawing the wave crests he had been imagining.

In 1896 he entered the Federal Polytechnic in Zurich. Before the new term began, Noah offered to organise his textbooks in advance and complete most of his exercises for him. Albert gladly accepted the first offer. He did not immediately refuse the second.

In the first lecture, the teacher began from a step he had never thought through properly. He looked down and searched the exercise book full of answers. Before he found the place, the teacher had moved on to the next line.

That evening he asked Noah to hide the last few lines of the exercises.

“Should I delete them?” the machine asked.

“No. I'll need to look if I can't do it.”

He pushed the notebook beneath the lamp and began at the first concealed line. Soon fresh corrections crowded the paper beside the answers the machine had given him.

Chapter Three Borrowed Notes

Grossmann kept very tidy notes. When Albert borrowed them, he often laid the notebook in the most prominent place on the desk and let Noah read it page by page.

“If you have Noah, why do you still need my notes?” Grossmann asked.

“It wasn't at today's lecture.”

“You could have been.”

Albert laughed and promised to attend the following week. The following week he spent another whole day elsewhere, reading.

One day Grossmann came to collect his notebook before Albert had finished copying it. Albert asked to keep it one more night. He would work through the difficult pages that evening, he said, then let Noah copy the rest.

“And what am I supposed to revise from tonight?”

Grossmann held out his hand. He had examinations to prepare for too. There was ink on his cuff that he had tried and failed to wash out. Albert did not ask him to wait again.

He handed back the notebook and attended the lecture the next day. He did not agree with everything the teacher said. Several times he wanted to leave, but stayed until the end. That evening Grossmann read his notes, laughed at the handwriting, then pointed to a passage and asked whether it was something the teacher had said or Albert's own objection.

Albert could no longer tell.

They checked it together and copied late into the night. As they left the room, Grossmann was still teasing him about his handwriting, saying that next time he might as well ask Noah to guess. Albert protested, then laughed himself. Just before they parted, Grossmann asked whether he would like to go for a walk at the weekend.

“Will you bring your notebook?” Albert asked.

“Come if you want. But don't borrow my notebook at the weekend.”

Albert attended lectures for a few days, then began missing them again. Noah could always supply the derivations he had missed; he felt there would be time to catch up. He kept putting off the subjects he disliked.

Once, Mileva looked at a solution he had brought, changed the example, and asked him to explain the third line. Albert talked for a long time, still repeating Noah's words. She waited until he fell silent, then pushed the paper back.

“When you come back, don't recite its second explanation to me.”

He was angry all evening. Only when he worked through it himself did he discover that her altered example failed to satisfy a condition required by the original solution. Noah had seen only the original problem. He was about to demand that the machine check for all such cases in future, then went to fetch the new diagram instead.

He switched the machine off. A few days later, during an experiment, he had to bring it back to the table.

He calculated three times using the relations Noah supplied. Each time he obtained the same result; the meter refused to agree. Grossmann checked the wiring. Mileva asked whether the room was at the same temperature as before. Albert found these details tiresome and asked the machine to calculate more precisely.

On the fourth attempt, the discrepancy remained.

Grossmann tried holding a terminal steady while Mileva took another reading and Albert wrote it down. At the fourth reading, Grossmann's grip loosened and the value jumped. They tried several more times and found the unreliable contact.

“The conditions I gave you were all correct,” Albert said when he got back.

“They did not describe the entire apparatus.”

He was about to argue when his fingers touched the blackened cloth in his pocket. He stopped. Cleaning the apparatus had left his sleeves so dirty that he had finally had to take off his jacket and wash it.

The clothesline hung beside the window. Noah continued to display the faultless calculation while water dripped from his cuffs. On the first page of the record, he drew a small diagram and added the connection he had left out.

Chapter Four Mileva's Notebook

Mileva's notebook was different from Grossmann's.

Grossmann liked to set each lecture out clearly. Mileva's margins were full of tiny writing. A line would lead from the top to the bottom, sometimes turning back to the previous page. Albert borrowed it once and complained that he could not find anything. She took it back and opened it at the page he had missed.

“You always start by looking for the answer.”

She was the only woman in their class. Some people could not remember her surname, but remembered that she came from Serbia. Others were surprised that she had come to study physics, as though every time she entered the classroom she ought to explain herself again. After enough questions, she sometimes merely nodded, opened her notebook, and waited for the lecture to begin.

One evening Mileva laid a problem in vibrations on the table. She had completed the assigned work, then changed the way one end of the string was fixed. She wanted to know at which step the original solution would fail. Albert read it quickly and said the result ought to be much the same. She handed him the pen and asked him to show exactly how much “much the same” was.

They began to argue. He asked Noah to decide. The machine offered three possible approaches. Partway through the second, she raised a hand to stop it.

“Don't choose for us yet.”

She drew a string on the paper and marked its ends. Albert saw his mistake at once and wanted to continue the derivation, but she was still pressing the pen hard over the two ends.

“Whenever you think you see something in the distance, you skip over this.”

“The distance is more interesting.”

“Then who's going to fasten the ends for you?”

They did not finish the problem that night. Someone downstairs was playing the piano, making the same mistake over and over. Albert hummed along and deliberately stopped at that point. At first Mileva ignored him. Then she laughed so hard she could not hold her pen. She hid her face in the crook of her elbow. When she had stopped laughing, she pushed the notebook a little nearer the middle of the table.

It was raining when they left. They stood in the doorway. Still thinking about the string, she caught his hand and traced on his palm the point she had not quite explained. He looked down, waited for her to finish, then closed his hand.

“Did you understand?”

“Yes. Draw it again.”

She looked up, saw him smiling, and smiled too. The rain did not stop. They squeezed beneath one umbrella. Only as they neared their lodgings did they notice that, by pushing it back and forth, they had both got their shoulders wet.

He loved evenings like that. One day there would be a teaching post, a room with two desks, and books they could read together when their work was done. He spoke in such detail that he might already have seen the room. Mileva added things of her own: the windows must face the light, and his books must not take up all the shelves.

The rent Noah calculated was higher than they had imagined. Albert said it would be nothing once they both earned salaries. Mileva still refused to cross out the second desk.

A young girl came to Mileva for help with mathematics. They went round the problem several times on paper, but the girl still did not understand. Albert sat at the other end of the table and could not resist interrupting. He spoke faster and faster. The girl stopped asking questions.

Mileva asked him to be quiet for a moment and began the first diagram again. She left the intermediate steps blank and waited for the girl to write them herself. When the final line appeared, the girl smiled and immediately brought out another, similar problem.

After she had gone, Albert said Mileva could always use Noah to make the explanations more complete.

“I want to see how she does it herself.” Mileva gathered up the paper. “Once you start talking, she's afraid to write anything.”

She walked faster than usual that day, saying she wanted to put problems like these into a little booklet. He helped invent a few. She crossed out the ones that were too difficult, and the ones that were merely showing off. They argued all the way home. Albert kept one problem until the following day, working out a way of asking it that he was certain she would like.

She read it and said that one would do.

A few days later the girl returned with a drawing of a garden. Mileva had asked her to use the same length of fence to enclose different shapes and compare how much could be planted inside. Instead of beginning with a calculation, the girl widened a long, narrow plot and drew two rows of vegetables. Then she erased one row, saying there had to be room for a path.

Albert was about to point out that the path had nothing to do with the problem, but Mileva had already moved beside her and was asking where the gate should go. Their discussion moved beyond the garden, and at last the girl explained the question of area she had not understood before. Mileva kept the drawing in her booklet and wrote beside it: Let her draw her own garden first.

As the final examinations approached, she slipped those sheets among her revision notes. The books piled higher; the booklet remained unfinished. She still remembered the girl asking, before she left, when she could come again.

In 1900 Albert received his diploma. Mileva failed her final examinations.

When the results came out, he waited outside for her with words of comfort already prepared. An examination meant little; those teachers did not like him either. After a sentence or two, Mileva asked whether he thought she had no reason to be upset.

“That's not what I mean.”

“Then don't explain just yet.”

He walked beside her for a while. A tram passed, and she stepped aside, still gripping the paper. He had his diploma in his pocket and did not know what to do with his hands. When the tram had gone, she was the first to move on.

She decided to try again.

That winter Mileva went back through her courses while Albert searched for work. They spent more time apart. Their letters still contained books and physics, but also lodgings, fares, and family matters that grew harder to explain. His mother disapproved of the relationship. Albert wrote passionately of their future. In her replies, Mileva still had to ask where they would meet next.

Noah prepared a revision schedule, which Albert copied into a letter. Mileva mentioned it only at the end of her reply: it was a good schedule, but he did not know how much time she actually had to study that week.

He asked the machine whether it could make another one to suit her circumstances.

“She would need to describe those circumstances.”

Albert picked up his pen again and began by asking how she had been.

Her next letter came late. He first looked to see how far she had got with her revision, then found a short passage about his search for a job. In his previous letter he had spoken very badly of himself, as though failing to find a position meant that even the books they had read together no longer counted. She returned instead to their last argument, set out a question of her own, and left a few lines for his answer.

Albert moved the job applications aside and began to think through her question. On the second page he remembered that she, too, was preparing for an examination she could not afford to fail again. At the end of his letter he asked whether she had been sleeping well. He had meant to write a great deal more about the future. This time he simply asked her to send him the problems she found most difficult.

Mileva failed the examination a second time. By then she was pregnant. She took her luggage and returned home. The room with two desks remained in their letters, described more briefly each time.

Albert wrote her a long letter and asked Noah to check its tone. He wanted her to know his feelings had not changed. The machine, however, made everything still uncertain sound securely settled. He reached the words “when everything is arranged” and suddenly asked when that would be.

Noah had no date.

He deleted the phrase, wrote down the next date when he would be free, and asked whether she could meet him. He read the letter twice, then sealed it.

Early in 1902 their daughter Lieserl was born. In another city, Albert read the news of the child, turned the letter over, then turned it back. He asked many questions. Some went into his

reply; others he spoke to Noah but did not write down. He still had no steady work. He still said they would soon be together.

Noah preserved some of that correspondence. Later, the records concerning the child broke off. The surviving letters could not establish what ultimately became of her. Noah's catalogue stopped there.

When Mileva eventually came to Bern, her trunk was not large. She unpacked her clothes, then took several books and old notebooks from the bottom. Albert reached to help. She shook her head and set them on the table herself.

He had found a job. His voice was light again as he spoke of their life ahead. Mileva listened, occasionally asking what they would do about marriage, how much the rent was, whether there was another chair at the table.

That evening he wanted to tell her about a problem he had been thinking over during the day. She said yes, washed her face, and sat down. He had only reached the second step when he noticed that her eyes were closed.

“Are you listening?”

“Yes. Tell me the second step again.”

He saw the shadows beneath her eyes and put the paper down, saying there would be plenty of time. Mileva caught his hand and asked him to stay a little longer.

She had imagined this evening many times on the journey. Now that she was here, the room was smaller than it had been in their letters. Her belongings were not yet put away, and she did not even have the strength to listen to one explanation all the way through. Looking at the paper on the table, she told him how much she had missed hearing him talk like this.

Albert did not resume at once. He sat beside her and let her lean against his shoulder. After a while she patted his hand.

“You can tell me now. Slowly.”

He went through the second step again. Halfway through, she asked a question he could not answer, and they both looked at the paper. Only when the lamp began to dim did she rise and point to her luggage.

“Don't put my notebooks at the bottom.”

Albert cleared a small space for them. Several books stood beside the lamp. The cover of one was worn through, but on its title page her own name was still clearly written.

Chapter Five The Patent Office Lamp

Before moving to Bern, Albert always looked at the end first when he opened a reply. The polite phrases were much the same. One letter sounded remarkably like a draft Noah had written for him. When he had finished reading, he turned it over and used the blank space to note the next address he would write to.

After graduation in 1900, everyone else's future gradually became clear. His did not. He kept several replies in a drawer; other letters went unanswered. At first he was too angry to sleep. Later there was no time for anger. The landlord had already asked twice whether he intended to stay the following month.

He tutored students, applied for jobs, and went back to mathematics he had once thought he understood. Noah listed ways to improve his prospects. It advised fewer unnecessary arguments and warned him against dismissing a correspondent's research too casually in a letter.

“You think they're right?”

“You have rejected their work before asking about the position.”

Albert crumpled the half-written letter, then smoothed it out. He revised it, deleting two offensive sentences. The final paragraph was too polite to sound like him, however he read it. He put one sentence back.

Grossmann continued to look out for vacancies and asked his family to help. In time there was news from the patent office in Bern.

Before going, Albert asked Noah whether this meant he could no longer do physics.

“The available evidence does not support that conclusion.”

“Can't you say something encouraging?”

Noah read out the salary.

He gave a short laugh, repeated the figure, and took out some paper. He subtracted rent and food. There was something left. He calculated it again and wrote to tell Mileva.

In 1902 Albert began work in Bern.

With his first regular pay, he settled several long-overdue bills, bought paper and a few household necessities, and set Noah on the table in his lodgings. His own manuscripts would have to wait until evening.

During the first few weeks he often came home too tired to talk. Unfamiliar terms had to be looked up; office rules had to be learned. One application appeared simple enough, but when he tried to put his assessment into words he revised it several times. At home he drew a clean sheet towards him to begin his own work. His pen hovered, and some unfinished detail from the day returned to him.

Noah offered to help read the applications. Albert prepared a description and asked the machine to summarise it. The summary was clear. Even the step he had deliberately omitted had been filled in.

He told the machine to remove the summary and went back to the drawings.

After a while, the assessments that had given him such headaches began to interest him. One ungainly design had a particularly clever mechanism at its heart. He followed the drawings, imagining the sequence of movements, and that evening used parts on his desk to work it through again. Mileva asked whether it was his invention. He said no, then explained the applicant's solution.

“You were smiling when you came home,” she said.

He was surprised. In his letters to friends, he had always described the job as something he was obliged to do.

Later they married, and their days acquired regular expenses. One evening another reply about a position arrived. It was brief and meant much the same as the others. He laid it face down at the corner of the table and told a funny story from the patent office over dinner. Mileva laughed with him. After the meal, she moved her chair beside his.

At last he said that sometimes he wondered whether everyone else had seen something he alone refused to admit.

“Seen what?”

“Perhaps examining drawings here is all I'm fit for.”

Mileva turned the letter over and pointed out that it said nothing about the problems he was working on. He said of course she would take his side.

“When have I ever pretended to agree with you about something I didn't understand?”

He remembered the string and had to smile.

She asked him to explain what had troubled him that day. The first attempt was confused, and she interrupted him halfway through. On the second, he stopped himself and got up to find paper. She moved the plates aside and wiped the table dry so that he could write there.

Near bedtime, he was still talking about the next problem. She covered the paper with her hand.

“Tomorrow evening, listen to me first for a while.”

“About what?”

“You'll find out tomorrow.”

He remembered on his way home the following day. When he opened the door, Mileva had already taken out her little booklet. The rather crumpled pages held the problems she meant to teach her pupils. He pulled up a chair without first asking what letters had arrived.

Patent applications often contained words such as “obviously,” “naturally,” and “undoubtedly.” He had to take them apart: how did the gear turn, how did the signal travel, what precisely linked one action to another? Some applicants wrote as though the machine already existed, while the crucial part of the drawing remained concealed behind a sentence.

Gradually it became a habit. He distrusted those words in his own manuscripts too.

There were fewer errors in the calculations Noah checked. Yet certain perfectly ordinary words grew less clear to Albert each time he read them.

“Simultaneous,” he said one day.

“Please specify the events.”

“Two clocks, both striking twelve.”

“In what locations?”

“One here. One at the other end of the street.”

“How do you compare them?”

He stopped writing, drew two clocks, and put a street between them.

Besso later joined the patent office. They often walked part of the way home together. As Albert grew excited, his pace would quicken until Besso caught his sleeve and asked him to explain a sentence he had passed over.

Once they stopped at a corner and argued while looking at a distant clock. Passers-by went round them. The clock moved on another minute.

Albert said someone could stand where both clocks were visible. Besso asked how long the light took to reach that person. Albert proposed sending a signal instead. Besso asked what he would do about the returning signal.

“Do you agree or don't you?”

Besso laughed and said he did not yet know what he was being asked to agree with.

They had passed the corner where they should have parted. Besso looked at the sky and turned back. Albert followed, so Besso stopped and pointed out their respective routes home. They set off along different streets, still calling back to each other. Only when they had to raise their voices did they finally wave goodbye.

At home Noah asked whether he wanted to record the discussion's conclusion.

“There isn't one today.”

“What is the principal disagreement?”

Albert draped his coat over the back of the chair and sat down at the table again. He remembered Besso's questions. When it came to his own answers, he could not carry on.

That evening he did not hurry to add another formula. He wrote down how the clocks should be synchronised, then read the description again, as though it were instructions for an apparatus someone actually had to build.

Chapter Six The Olympia Academy

In 1902, newly arrived in Bern, Albert was still earning money by giving lessons. Solovine first came to him looking for a physics tutor.

He had seen Albert's advertisement and arrived with several questions, expecting to pay for his time. They sat down, and before long the appointed hour was over. Solovine glanced at the clock, but Albert was still asking about a book he had mentioned.

“Does this count as part of the lesson?” Solovine asked.

Albert hesitated. He needed money, but he had not read the philosophy book Solovine was describing.

“Not today.”

In time they stopped calling them lessons. When Habicht joined them, the three gave themselves a grand name: the Olympia Academy.

They still met in Albert's lodgings. Whoever arrived first cleared the table, and the Academy was open. Bread, cheese, and sausage stood beside the books. Once wiped clean, the knife they used to divide the food sometimes held down a curling page. Solovine read philosophy; Habicht brought mathematical problems; Albert could turn almost any remark towards the physics that was troubling him.

They teased one another too. If someone grew too solemn, the other two called him President. If the President could not explain himself, he was dismissed on the spot.

When Solovine first heard a voice behind the bookshelf, he thought there was another lodger next door.

“There's nobody there,” Albert said.

They brought the grey box to the table. Habicht walked round it and asked why a device like this had never been patented. Albert said he could not explain how it had been made, or how every part inside it worked.

“And yet you're so well qualified to find fault with everyone else's work all day.”

Habicht reached out and tapped the casing. Noah informed him that physical impact was unnecessary to establish a connection.

The three of them laughed for a long time.

When Habicht had stopped laughing, he still wanted to test it. He brought a sheet of calculations he had checked in advance and deliberately included an ambiguous problem. Noah answered the first almost instantly. On the second, it selected one interpretation and continued explaining.

“It can pretend we've made ourselves clear too,” Solovine said.

“When you were talking about ‘experience’ just now, you didn't ask whether we meant the same thing,” Albert replied at once.

They had meant to examine the machine, but were soon arguing with each other. Habicht tapped the table with his pencil and demanded that they finish his experiment first. He recorded the results, turned the sheet over, and sketched the dimensions of the box. At the connections he stopped measuring and said he would bring another meter next time.

After that, Solovine sometimes broke into a discussion to ask whether anyone had actually read the page they were criticising. All three would look for the book. The machine, too, had to identify the passage it was quoting. Once everything had been found, the most confident speaker was often the first to laugh.

At first Solovine liked Noah very much. Whichever book he asked about, it could supply several passages. Later he complained that it interrupted: they had barely reached a contentious sentence before the machine had recited all the competing interpretations, leaving them no time to argue.

That evening they were reading Hume. Solovine finished a short passage, looked up, and asked Albert what entitled him to believe that one thing caused another.

“Because I can write out the process.”

“How many times must it repeat before you can say the next time must be the same?”

Albert wanted to use electromagnetic theory as an example, but Habicht moved the lamp towards them.

“Let's begin with this lamp.”

Noah began explaining the problem of induction. Solovine stopped it, pulled the lamp closer, and read the passage again. At the word “necessary,” Albert interrupted. The three began arguing once more.

Albert disliked some of what he heard. He thought Solovine carried the questions too far, to a point where no experiment could even begin. Yet the next day, examining a drawing, he underlined the words “necessarily produces” and held his pen there for a long time.

With Mach the argument took another direction. With Poincaré they quarrelled over measurement and convention. Habicht insisted that they write down what each meant by “time.” Otherwise, by the time the word had travelled round the table, it had become three different things.

“Four,” Noah said.

Habicht drew the paper towards him and added another column.

Some evenings they scarcely turned a page. The sentences they had read circled back on themselves until everyone was repeating himself. Habicht wanted a walk. Albert insisted there was one last point. That point took another half-hour, and Solovine had already put on his hat.

At last they went outside. The night air in Bern was cold. Conversation abruptly thinned as they walked with their hands deep in their pockets. They could not see the water beneath the bridge, only hear its unbroken sound.

Albert brought up the book again. Solovine said the President had adjourned the meeting. Habicht was feeling the brim of his hat. A man who had just passed was wearing exactly the

same one, he said, yet somehow looked more respectable. Solovine glanced at him. “You can't blame the hat for that.”

Back at home, Noah asked whether it should add to the record of the discussion. Albert said there had been no more discussion. He was still smiling when he lay down.

Noah's edited record ran to only a few pages. It omitted most of the evening's digressions and jokes. Albert read quickly, reached the last page, and still had not found the part he wanted to hear.

He asked it to restore a passage from the middle.

“This passage has no conclusion.”

“What did Habicht say that made Solovine drop his bread?”

“It has no direct bearing on the current proposition.”

“I'd like to hear it again.”

The voices resumed. Albert heard himself interrupting everyone, half a sentence carried away by another, someone insisting on an immediate objection with his mouth full. He laughed. Later he caught up a pen and wrote down one of Solovine's questions. He had been so busy refuting it that night that he had not properly heard it.

After Hans Albert was born in 1904, the Academy acquired one explicit rule: do not wake the baby.

Mileva sometimes sat at the table for a while. She knew the books they were reading and would pause to challenge something. When the baby cried, she was usually the first to rise. By the time she returned, they had turned several pages. To rejoin the conversation, she had first to ask where they had got to.

Once, she did not get up.

Albert was in full flow. His voice faltered. She looked at him and pointed to the next room. Only then did he put down his book. When he came back carrying the child, the other two had altered his explanation beyond recognition.

“Where did I start going wrong?” he asked.

Mileva turned back two pages and pushed the book towards him.

“You try picking it up from here.”

He wanted to protest, but the baby pushed a hand inside his collar. He sat down and asked them to go more slowly.

They finished early that evening. Solovine helped clear the table; Habicht took away the books that needed returning. Albert handed back the sleeping child and tried to explain that he had lost track of time. Mileva took the baby and said she knew.

He said he could tell them not to come the next day. She looked up and asked why.

“You're tired.”

“Then you could hold him a little longer.”

She still wanted to hear them discuss the book. The child had been restless all afternoon. She had opened the page and never managed to read on. Could they take turns holding him, she asked, so that she could finish what she wanted to say before taking him back?

He took the child and walked slowly round the room. She returned to the page and began again. When she could not concentrate, she looked up at him. The baby was tiny, and Albert still held him awkwardly. Passing the table, he bent to kiss her forehead and almost knocked a cup over with the baby's foot.

She caught the cup and laughed without a sound.

Later Habicht left Bern, and Solovine too was preparing to go elsewhere. At their last few meetings they discussed books as usual, remembering halfway through to ask whether the luggage was packed and when the train would leave. No one announced the Academy's end. Some evenings Albert still cut two extra portions of bread and only remembered, as he brought them to the table, that his friends had gone.

He wrote to tell them of new questions. Noah asked whether he should enclose summaries of all their discussions.

“Not yet.”

Albert crossed out the polite opening and began by asking Habicht whether he had changed his hat. Only then did he describe what he had been doing.

He wrote late into the night. As he signed, he realised that even the serious passages sounded as though he were arguing with a friend.

Chapter Seven 1905

In 1905, it was almost impossible to tell one pile of paper on the table from another.

Calculations of molecular dimensions crowded questions about light. The motion of tiny particles in a liquid occupied one sheet; two clocks occupied another. He would explain a problem to Besso and think of a different one halfway through, only to return home and get stuck on an inconspicuous coefficient.

Noah sorted the work into categories. Albert kept mixing it up again.

Before clearing the table, Mileva asked which papers could be moved. Sometimes Albert said none, then moved them all to the other side himself a moment later. She would weigh down one pile with an empty cup, so that after dinner he would not ask where the things that had just been here had gone.

Once she pointed to a crossed-out passage and asked what, exactly, he had changed. He began explaining the definition of simultaneity, speaking faster and faster. When he finished, she asked how the two observers would each make an actual check. He pulled a sheet towards him. Halfway through the drawing, the baby knocked over a cup.

They wiped the table first. When they returned, she had written the two versions of her question in the margin. Albert went on drawing. Mileva sat for a while, then rose to finish the housework. Noah preserved the conversation. In the later hours of the night, only his voice remained.

Some nights Mileva had already gone to bed while he sat at the table, bringing each newly written passage closer to his eyes. The child woke again. Albert had barely stood up before Mileva was beside the bed.

“I'll do it tonight,” she said.

The next morning, dressing the baby, she fastened two buttons wrongly. Albert pointed it out. She looked down for a while, then suddenly began to cry. At a loss, he started saying that the previous night's work had needed only a little more. She shook her head and asked him not to explain yet.

He took the baby. She sat on the edge of the bed with her face in her hands. After a long time she looked up and said she was simply so tired.

When he returned that day, he meant to resume work immediately. Instead he put down his bag and took the child for a walk. The lamp in the room remained unlit. By the time Mileva woke, he had heated water but could not find the baby's clothes. From the doorway she told him which drawer, then asked whether he had understood that last little thing from the night before.

He said he had not. She nodded, poured water into a cup, and sat beside him. She asked where he had put the paper. He found it, then asked whether she had slept enough.

While he was writing the paper on Brownian motion, Noah quickly calculated trajectories under different conditions. Points jumped about on the screen like fine dust floating in a shaft of

light. Albert wanted more than this. He needed a measurable relationship if there was to be a way of testing whether the unseen molecules existed.

“Give me more trajectories,” he said at first.

Later he said, “Enough. List the things we can measure.”

Noah once gave him an almost flat curve. Many points moved left and right; their displacements cancelled one another, positive against negative. It looked as though nothing had happened.

Albert enlarged the display. One point had travelled far, another in the opposite direction. He asked the machine to stop there and traced several of the paths with his pen.

“They haven't stayed where they were.”

“The mean displacement is close to zero.”

“I'm not asking where they're all going together.”

He changed the way he recorded how far the particles had travelled, so that movements in opposite directions no longer erased each other. Each particle still wandered erratically. Taken over many movements, however, the measure of how far they spread bore a stable relation to the observation time.

That day he superimposed the results for different observation periods. Noah recalculated again and again; Albert kept changing the quantity he was asking about. The sounds outside faded and returned. Only when the sky grew light did he notice that his cuff had been resting on wet ink.

Albert suspected that, in certain phenomena, light's energy might need to be treated as discrete portions. Yet there was the success of electromagnetic waves. Noah calculated along both paths. The further it went, the more evident the disagreement became.

“Which account is true?” Albert asked.

“You need to specify how each is to be tested.”

He did not write that down. Both derivations lay on the table. He pushed one farther away, then later drew it back.

The manuscript on special relativity gradually took shape. The arrangements for synchronising clocks, the outward and return journeys of signals, the motion of observers: all had to be stated afresh. After checking the calculations, he went back through the text sentence by sentence.

He spent a long time on one page, then struck it out. To preserve a single “now” for every observer, his explanation grew more elaborate and acquired more exceptions. Noah could still make the additions consistent with one another, but only if he accepted further untested assumptions.

Albert looked at the page and thought of parts concealed behind “naturally” in patent drawings. He covered it and began again with two actual events. Noah asked whether he wished to retain the earlier requirement.

“Not for now.”

He went to fetch more water. On returning, he uncovered the page and reread his requirement. Giving it up would make the derivation much simpler. Still uneasy, he asked Noah to recalculate from the simplest case.

A few clocks, a few exchanges of signals: the differences in the results were hard to accept. For another observer in relative motion, two events previously simultaneous might occur in sequence. He checked every step again before he would keep that result.

Sometimes it happened the other way round. He insisted on an explanation, and Noah supplied a case in which it failed. He changed the example three times and still could not obtain the result he wanted. At last he tore off the whole passage.

“You win today.”

“This problem has no competitive metric.”

“You could have a sense of humour occasionally.”

He spread the torn sheet beside the bed. On waking the next morning, he read it again.

The manuscripts went out one after another. On some mornings he glanced an extra time at the letterbox before hurrying to the patent office. At night he continued thinking about the relation between energy and mass. Unwashed cups remained on the table.

Before sending one manuscript, he stood a long time in the doorway, then suddenly turned to Mileva. What if no one read this one properly either?

She was holding the baby. At his words she came over, freed one hand, and smoothed the curled corner of the envelope.

“Then come home and tell me.”

“You've heard it already.”

“Not all of it.”

He looked down at her and touched the child's face, then hers. She turned her cheek into his palm. After a moment she reminded him that the post office was about to close.

One day Mileva came home to find him carrying the baby round the room, murmuring something. She thought he was soothing the child. Only when she came closer did she hear a newly revised sentence. The baby was sleeping soundly, his face against Albert's shoulder, leaving a small damp patch on his shirt.

She reached to take him. Albert shook his head and said in a moment. The sentence was still not right, but he had learned to pause slightly as he rounded the table, so that the baby's feet would not strike the back of a chair.

Noah did not file that recording with the papers. It saved it under the day's date.

Chapter Eight A Letter from Berlin

When the letter from Berlin arrived, Albert looked first at the signature.

He had seen Planck's name many times in journals. At first he feared the letter contained no more than a few courteous phrases. Then he began the body of it and forgot to sit down.

It asked about a detail in the relativity manuscript. He returned to the beginning to make certain he had missed nothing. Planck had written to discuss that very step.

Albert took out his manuscript and marked the place.

“He read it all,” he told Noah.

Noah stated the number of pages in the letter. Albert ignored it and repeated, “He read it all.”

He sat down and began a long reply.

Halfway through, he realised he no longer needed the introduction he had copied at the top. He had planned to explain his training, how he had come to the patent office, and why he continued his research. Planck had asked about none of that. He had asked about the derivation.

Albert crossed out the introduction and returned to the condition Planck had marked.

The next evening Noah checked his quotations. Albert asked it instead to read one question from the letter again. He, too, had spent a long time at that step. Now someone else had stopped there.

He wrote too excitedly. The first draft explained several things Planck had not asked about. Noah suggested removing two paragraphs. Albert read them, deleted one, kept the other, and toned down several sentences that sounded like declarations of victory. Planck's letter lay beside him. From time to time he picked it up and reread it.

That evening Mileva asked who had written. He said the name twice.

“I heard you,” she said, still holding out her hand.

She took the letter and looked first at the questions. At one point she returned to the beginning. Albert stood beside her, urging her to read on. She told him to be quiet. When she finished, she looked up at him. Her lips moved, but no words came.

He had prepared so much to say. At the sight of her, all he said was, “You see.”

She nodded, put the letter on the table, and held him. Water was boiling in the kitchen. They both heard it; neither went at once. Albert pressed his face to her hair, still gripping the unfinished reply.

“He asked about this.” She freed one hand and pointed to a line.

They read it together again. Only afterwards did she ask whether this meant he could soon change jobs.

“We can't say that yet.”

She was a little disappointed, but found him fresh paper. As she handed it over, she made him promise to go to bed when he had finished: he had work the next day. He agreed, caught her hand, and asked her to sit a little longer.

The next morning Albert arrived at the patent office on time as usual. After assessing an application, he would suddenly recall one of the letter's questions and smile without knowing it. If a colleague asked, he had to begin with the two clocks again.

As the correspondence grew, Albert discovered that Planck agreed with him on some matters and would not yield an inch on others.

On relativity, Planck was willing to take time over the new theory's implications and bring it into serious scientific discussion. Light quanta were another matter. He had himself introduced the quantisation of energy in his work on radiation, but this had not led him to accept Albert's bold proposal about light itself. To Planck, that further step still lacked sufficient grounds.

Albert resisted. He wrote their arguments side by side and had Noah check them, then demanded another kind of check. The machine could identify their respective premises. It could not remove the physical disagreement for them.

“He can accept relativity. Why does he always think I've gone too far when it comes to light quanta?”

“The two pieces of work require their own evidence.”

He knew that. Even so, he could not help finding Planck's earlier letters and rereading the words of agreement.

Once, while he was preparing a reply, Noah condensed one of Planck's objections into a few sentences. Albert found it suspiciously easy to refute and asked for the original.

The machine had omitted a qualification. Restoring it exposed a gap in his half-page rebuttal.

He laid the half-written page aside.

“Keep that sentence in future.”

“The summary will be longer.”

“His question is that long.”

Years later, organising the archive from this period, Noah classified the correspondence as supportive, opposed, or undecided. Planck's name appeared in two columns. Albert looked at them, remembered the table in Bern, and reunited the categories beneath the same person's name.

In 1909 Albert went to a scientific meeting in Salzburg. For the first time he stood before an international audience of that kind.

Before he left, Noah marked several points in his lecture that might be challenged. He revised some and took the manuscript with him. As he spoke about radiation, someone in the audience turned back a page of notes. Albert slowed down and supplied an assumption he had just passed over. Several listeners were still shaking their heads.

After the session he had more to say and followed Planck towards the door.

He spoke too quickly. Planck reached out and held down the page Albert was about to turn.

“This step.”

Albert explained it again. Planck listened, but still saw no reason to accept so radical an account. One by one he brought up phenomena the wave theory already explained.

Albert grew impatient. He drew an example. Planck waited until he had finished and asked how the same account would handle another case. They soon ran out of paper and turned the sheet over. Someone standing nearby, waiting to leave with them, waited a little longer and finally went alone.

When they parted, neither had convinced the other. Planck wrote down a paper he had mentioned and asked Albert to read it. Albert asked him to reconsider his own question. Before putting away his pen, the older man checked the passage Albert had marked.

At his lodgings, Albert took off his coat and asked Noah to find the work Planck had recommended. The machine asked whether the meeting had met his expectations. Albert turned the pages and drank some water.

“I thought it would be easier to make myself clear in person.”

“Easier than in correspondence?”

“In a letter I can always keep writing.” He pointed to the manuscript. “Today Planck wouldn't let me skip that step.”

He spread out his lecture and added several lines to the page they had argued over most. Only then did he pack what he needed for the following day.

That autumn he left the patent office to teach in Zurich. While clearing his belongings, he found several old replies to job applications behind a drawer. He crouched and picked them up one by one. He could still remember the weather outside when one of them had arrived.

Noah asked whether the files should be reorganised to reflect his new appointment.

He laid Planck's letter beside the others. After a while he took it out again and slipped it into a pile of scientific questions he had yet to answer.

The new job excited and unsettled him. Preparing his first lectures, he repeatedly found himself halfway through an explanation before remembering that the students might not know the steps he had omitted in his head. Noah displayed the notes beside him. He split an overcrowded page in two, made space, and drew a diagram he would once have thought unnecessary.

Chapter Nine General Relativity

Several years after leaving the patent office, Albert still kept his early thoughts on gravity at the front of his manuscript.

He had imagined a person in free fall. If that person let go of an object during the brief descent, what would happen beside him? The thought brought gravity and motion close together. It did not, by itself, become a complete theory.

He trusted the intuition without knowing how far it could take him. Whenever he tried to turn the scene in his head into an argument someone else could examine, he stopped at a new difficulty. Words that came easily in conversation proved, on paper, to have no definition he knew how to give.

Noah could assist with the derivations and display structures more complicated than the mathematics he knew. He learned some and tried applying them to his problem. Something always remained unclear.

After his return to Zurich in 1912, Grossmann sat across from him again. Albert brought many pages, along with lessons left unlearned years before.

“First you borrowed my notes. Now the entire mathematics department?” Grossmann said.

Even as he spoke, Grossmann cleared half his desk and took a page of calculations. Which symbols stood for things that could be measured? Albert pointed them out one by one. As he went on, his voice slowed. He knew what he wanted to explain, but not which mathematical object could carry the meaning.

They drew several maps. The same place could be marked with different coordinates. The numbers on the paper changed, but the thing being described must not become something else. Noah could perform the transformations easily. Albert kept asking how to recognise the original physical problem in the result.

Once Grossmann stopped and told him to begin again from the previous page. Albert could almost see the next step and did not want to go back. He all but begged his friend to let him finish. Grossmann put down his pencil and waited with empty hands.

By the third sentence, Albert was stuck.

Only the rustling of paper could be heard. Grossmann said nothing and found the previous page for him. The next morning Albert arrived early and began by working through the three sentences that had stopped him.

Some geometrical expressions were beautiful but had no clear physical meaning. Some approximations worked well but imposed too many restrictions when extended. Noah continued producing candidates. Albert watched the stack grow, wanting it to stop, afraid he might miss something.

They rejected one direction for reasons that seemed sufficient at the time. Noah recorded them point by point. For months no one opened the file again.

In 1913 Planck and Nernst came to Zurich to invite him to Berlin.

The position, the research conditions, and the people he could work with appealed to Albert. Planck had helped secure the invitation, though his recommendation still expressed reservations about light quanta.

Albert wanted another argument about light quanta. Planck was already discussing how his work in Berlin could be arranged.

When they turned to the gravity research, Albert set out the difficulties on paper as well. At the unresolved part he grew impatient with himself and tried to follow it with a more promising result. Planck asked him to stop there.

“How much longer will this part take?”

Albert wanted to give a deadline that would not disappoint him. In the end he said he did not know.

Planck nodded and asked what conditions he needed for his work. The unfinished pages remained spread across the table while they went on discussing the arrangements. The deadline Albert had prepared never went into them.

When Mileva heard Berlin mentioned, her first question was about the children.

They had already moved between several cities. Hans Albert was growing up; Eduard was still small. Albert spoke quickly, as though finding a house would allow life to move with the research. Her questions came more slowly. Who would look after the children? Who would contact the schools? Whom did she know there?

Other things between them did not appear on his sheet of paper. He was growing closer to Elsa, and Mileva knew. When they quarrelled, he sometimes withdrew into his work. There, at least, he felt he could still take an expression a little further.

Noah had prepared a moving plan for him. It covered every item and calculated the costs. Mileva read two lines and put it down.

“Everything you need is here.”

Albert said they could add more.

He waited with his pen. Mileva did not look at the list again.

One evening she spread the children's belongings on the bed. Clothes they had outgrown went in one pile, things they could still wear in another. She asked Albert to look at a broken toy. He agreed, still holding a letter. Much later she took the toy away from in front of him.

Only then did he look up and say he could mend it now.

“I've packed it.”

He stood a moment, then reached for the lid of the trunk. She did not stop him. Before fastening the catches, they tucked in the clothes sticking out, their movements as familiar as on so many previous moves. Then she turned to another trunk. He meant to ask whether there was anything else he could do. What he spoke of instead was the unanswered letter.

She did not reply. In the trunk lay a stack of manuscripts he had been using. She had wrapped them in cloth and placed them where he could reach them during the journey. He noticed and thanked her. Her hand remained on the rim.

“I know you need to go to Berlin,” she said at last.

He began to explain what it could offer. She shook her head, adjusted the cloth beneath the manuscripts, and said they could discuss the children's schools tomorrow. There was one thing she wanted to ask tonight.

Once they reached Berlin, did he still want to share his life with her?

Albert looked at the trunk and said of course. He answered quickly, then went back to discussing the house. She lowered her head and tucked in the corner of the cloth. She did not ask again.

They arrived in Berlin in 1914. Before long, living together became impossible. Mileva returned to Zurich with the two boys. Before she left, Albert found one of her early notebooks among a stack of books. He recognised its cover. He meant to hand it to her separately, but in the end placed it in the trunk she had indicated.

Once the rooms were quiet, he could work for long stretches. Noah calculated a new daily schedule. Albert agreed, then stopped it halfway through.

The other side of the table was empty. By habit he gathered his scattered papers to leave room there. Only when he had finished did he stop.

Letters to the children now had to be written separately; money to send them had to be calculated separately. Sometimes Mileva wrote only of practical difficulties. Her letters irritated him. Reading them again later, he found that the things she described were indeed still unresolved.

In 1915 Albert returned to the gravity calculations he and Grossmann had rejected. The old directory marked them as failing the requirements. He asked Noah, “Why did you think this path was wrong too?”

“I retained the conditions you then required and excluded the proposal on that basis.”

“You didn't ask whether those conditions could be abandoned.”

“Neither did you.”

He copied the old requirements from the screen and examined them one by one.

“Record this. If you find me refusing to re-examine my premises, say so.”

Noah kept the instruction.

After repeated revisions, the calculation of the anomaly in Mercury's orbit finally agreed with observation. Albert called no one. He stood at the desk for a while, then recalculated the crucial steps.

The result on the screen did not change.

Noah asked whether it should begin the next test. He shook his head and asked it to leave this result on the screen. He sat down again and pressed his palms to the cold wood of the desk.

The heavily revised pages looked beautiful to him. Falling bodies on Earth and the motion of distant planets had come together within one description. Those abstract relations had accounted for so minute a deviation in Mercury's orbit. He looked at the observed value once more, his hands still on the desk.

Later he found the notebook he had used at twelve. A loose page had almost come free. The pencil on its front was faint, but he could still read: What is the origin of gravity? On the back were the two crooked dots he had drawn beside his compass.

At the time, he had always wanted to find some thing between objects that passed the influence along. Now he had to reconsider how distance was measured, how clocks ran, and how the motion of bodies related to the structure of spacetime. The words on the page had not changed. The question was no longer what he had understood as a boy.

Noah asked whether it should mark the question as solved.

He laid the new Mercury result beside the old page and said they should first record precisely what had been understood that day. He still had no answer he could guarantee about the deeper cause.

Someone passed outside, footsteps fading into the distance. He wanted to tell someone. He thought of several names, then of where those people were now. At last he drew the paper closer and darkened a mark that might easily be misread.

Albert took out his unfinished letter to the children. There were no formulas on that page. He read it, crossed out a sentence explaining how busy he was, and continued.

Grossmann's name remained on the gravity paper they had written together. Whenever Noah opened the record of that work, it also opened the pages Albert had once set aside.

Chapter Ten After Fame

In 1919 Albert and Mileva divorced. That same year, observations of a solar eclipse brought his theory to the world's attention.

The newspapers printed his name in large type. Mileva was still writing about money owed; the children were waiting for replies; old household accounts remained unsettled. Noah filed the papers by date. A letter asking for payment was followed by a newspaper bearing his photograph.

Journalists liked to ask about a moment: on what day, where, how had the idea suddenly come? He tried to describe the preceding years, Grossmann, the things he had changed and changed back. They nodded attentively. When he paused, they asked whether he could make it shorter.

In print, those years became a few sentences. Reading them, Albert felt that some passages were not exactly wrong. He simply could not remember having lived so smoothly.

Later, when asked for the same story, he too left much of it out. He was not always in the mood to explain from the beginning, and the familiar laughter was pleasant. Back in his room, the unfinished problem remained beside the desk. He sat down and resumed correcting a line he had got wrong that morning.

After he became famous, Albert received many letters from strangers. Noah sorted them by origin and subject. Gradually the wood of the desktop reappeared. One clumsily written technical objection always came last.

He drew it out and read it. Its conclusion was wrong and its reasoning was wrong, but one part of the question deserved a better explanation. He looked at the sender's work address and asked whether, using this method, Planck would have read him first all those years ago.

Noah could not determine that. Albert looked at the unopened pile and found no immediate solution either.

That evening he wrote two pages in reply and kept the original to check his revised explanation. A few days later the table was covered in new letters. Sometimes he slipped one beneath a manuscript before finishing it and found it much later. By the time his reply went out, the sender had moved.

In Berlin he sometimes went to Planck's house to play chamber music. The older man sat at the piano, turned a page of the score, and waited for him to settle his bow. Sometimes an argument left unfinished during the day entered the room with them. After a few remarks, they saw the others had their music ready and put it aside.

Albert came in half a beat early. When they stopped, Planck's hands were still on the keys. Together they found the bar in the score. On the next attempt Albert slowed a little.

At the end of the piece, Planck looked up. Before he could speak, Albert asked whether they might agree that this performance had never happened.

Walking home, Albert hummed a short phrase. Noah asked whether it should find the score. He said he knew it. Only on reaching home did he notice that the argument which had occupied him all day no longer seemed quite so intolerable.

The growth of quantum theory brought new disputes. In the winter of 1926, Albert wrote to Born, acknowledging the theory's achievements while refusing to believe that probability provided nature's final explanation.

“God does not play dice,” he told Noah. His letter said much the same.

Noah left Born's paper on the desk. Albert did not ask it to remove it. He read it again and wrote questions in the margins. The next morning one of them seemed to miss the point, and he crossed it out.

At the following year's Solvay Conference, he met Bohr. Albert sketched an apparatus in which an electron passed through a small opening and reached a screen. He kept altering it, searching for a gap in quantum theory. Bohr spoke slowly, though his finger moved quickly across the drawing. Sometimes, as Albert began to interrupt, Bohr put a hand over the paper and asked him to hear the measurement conditions first.

“With many electrons, the distribution comes out correctly.” Albert tapped the screen in his diagram. “But this one before us—why does it arrive exactly here?”

Bohr returned his hand to the drawing and asked how he proposed to put that question to an experiment. Albert thought he had made himself clear. After listening to Bohr, he found that they were still stopping at different places. He would not concede that an account of an individual event must end at probability.

In a 1929 interview, asked about imagination and knowledge, Albert said, “Imagination is more important than knowledge. Knowledge is limited. Imagination encircles the world.”

As Noah archived the interview, it asked whether to link the passage to his early record of chasing light. Albert agreed. Then he found several pages of calculations repeatedly revised with Grossmann and asked for those to be included too. One had been crumpled. When unfolded, a crease crossed the point at which he had once been unable to proceed.

In 1930 they returned to Brussels. Albert brought a box.

It was drawn on paper, hanging beneath a balance. The box contained light. A clock controlled the opening and closing of a shutter, allowing a single photon to escape. Weighing the box before and after would reveal the energy the photon had carried away; the clock would identify the instant it escaped. Both measurements appeared capable of arbitrary precision. If the argument held, it would evade the uncertainty restriction involving energy and time.

Bohr did not answer immediately. He brought the paper closer and asked about several details. Albert supplied them, then laid his pen across the page beneath the box and waited.

When Bohr spoke again, he pointed first to the clock.

The box's position during weighing had an uncertainty. According to general relativity, the rate of a clock depended on its position in the gravitational field. Taken together, these effects meant that Albert could not preserve an arbitrarily precise time for the clock while weighing the energy with arbitrary precision.

“We have to use your relativity here too,” Bohr said.

Albert picked up his pen and recalculated the lines beside the balance. The first time he paused, Bohr thought he was about to speak. Albert merely turned the sheet over and calculated again.

“This method doesn't work,” he said.

At last Bohr moved his chair back. Albert was still looking at the drawing. After a while he raised his head and asked another question. Bohr had to sit forward again.

That evening Noah laid out their arguments side by side. Albert corrected his omission. The original box drawing went beneath the other papers. He did not try again to use it to overturn the uncertainty restriction.

He still wanted to know why a theory that calculated so well could not give a fuller account of a particular event. Relativity had led him to believe that, under appropriate conditions, changes in matter and spacetime could be traced through deterministic laws. Quantum mechanics, too, possessed rigorous laws of evolution. Yet for particular measurement outcomes it could often give only probabilities. Did the difference arise from nature itself, or from the limits of the existing description?

Noah asked whether deterministic laws should remain a requirement for a new theory.

“Leave it for now,” he said. Then he added, “Mark it as unproved.”

Returning to Bohr's argument, he noticed a word he had copied so carelessly it might be mistaken for another. He rewrote it clearly.

That year, in a letter to a friend, he had made a wry remark: “To punish me for my contempt for authority, fate has made me an authority myself.”

One correspondent wrote about the Brussels dispute, saying that as long as Albert continued to object, quantum theory ought not to be accepted too readily. The letter was very respectful. It said nothing about the clock in the box.

Albert put it aside and asked Noah to find Bohr's argument. He still disagreed with some of Bohr's interpretations, but the failure of the photon box had been clearly established.

“Send him this too.”

Antisemitism in Germany grew louder. Threats began to arrive among the letters. People came to say goodbye without yet knowing where they could go, only that they must leave. During his travels Albert kept hearing from old friends. Some letters could reach him only through a third person.

A teacher with whom he had once exchanged measurement records sent two slim sets of lecture notes. The older edition contained an acknowledgement. The new one removed the name of a Jewish collaborator while retaining the corrected readings unchanged. The teacher wrote that he had objected and been told there was no need to print the old edition again. Now he was no longer permitted to teach the course either.

Albert opened both booklets at the same page. He did not know the person whose name had been removed.

“Can you find him?”

Noah had only an address in the lecture notes. It was old, and the teacher who had written did not know where his collaborator now was.

At the end of the appeal, the teacher listed several things: the subjects he could teach, the instruments he could repair, whether anyone might offer even six months' work to begin with. His wife had stayed behind to deal with the house. He had not given her address and asked that any reply be passed first to the person forwarding his mail.

Albert drafted a recommendation. At “trustworthy,” he stopped, found the measurement he had actually read, and replaced the phrase with a few specific statements. He wrote separately to two friends, asking about vacancies.

One replied that there was no money. The other was willing to meet the man but needed copies of his documents first. The teacher answered several weeks later, thanking Albert and asking him to confirm whether his wife could come too. The confident phrases of the original recommendation could not decide how they would cross a border.

Albert eventually settled in Princeton. Noah travelled with him, acquiring a long dent in its casing. Some papers were guarded more carefully than clothes; some possessions were still lost. Both sets of lecture notes survived. He filed the correspondence about finding work behind them, apart from the arguments over relativity.

After that, one old friend remained silent for a long time. Albert repeatedly asked Noah to look for a new address. The record always ended at the same place, so he tried another spelling of the name. The door hinge in his new home creaked. Even the trees outside were unfamiliar. When the postman came, he looked first for handwriting he knew.

One evening he played an old recording of the Academy and heard his younger self urging the others not to digress. He began to laugh, then sat listening while they finished all the words that remained of that night.

In 1935 he wrote a paper with Podolsky and Rosen, continuing to ask whether quantum mechanics was complete. After two systems interacted and separated over a great distance, a measurement of one could still allow certain measurement outcomes for the other to be predicted. If the distant system was not disturbed by what happened here, why could its properties not be described independently?

Bohr's reply came quickly. He argued that what could be predicted about the distant system still depended on the measurement conditions chosen here. Albert would not accept this answer. Noah asked whether he wished to write another letter. He had already taken out paper. After a moment, however, he asked the machine to read Bohr's article from the beginning.

In his new room he continued searching for a unified account. One day, reorganising the old archive, Noah found that the folder marked “Set Aside for Now” had grown larger than the entire original collection.

Albert would not let it delete anything.

Chapter Eleven The Last Equation

In 1949 Albert turned seventy.

Noah's casing had been replaced several times. Those who maintained it could not make sense of the internal construction. They could only replace connections and repair the power supply. It still worked, but small amounts of data were beginning to disappear. Some voices survived; others existed only as text. The right side of one old photograph had become permanently grey.

The machine could fill the grey area with a generated image. Once it produced a picture that appeared complete: window, chair, and the edge of a person's coat all joined naturally. Albert looked at it for a long time, then asked whether the chair had actually been there.

“It was generated from the surrounding content.”

He asked Noah to display the two pictures side by side. The chair looked familiar. He thought and thought, unable to remember whether he had sat in it.

“Keep the original.”

Sound presented the same problem. In an old musical recording, he had played a wrong note. The machine could correct it precisely. He listened once and found it beautiful, but would not let the corrected version replace the original. Just after the wrong note someone had given a short laugh. Once it was corrected, the laughter had lost its cause.

One day, reading an old directory, Noah asked whether several duplicate records should be merged. Albert heard Solovine's name and told it to stop there.

One recording still played. Someone was arguing over a page of a book. Someone else laughed too loudly, obscuring the rest of the sentence. The old man turned the sound up, then down, and finally asked to hear it from the beginning.

“Habicht was always saying I interrupted.”

On the recording, the young Albert was taking over someone else's unfinished sentence.

He listened, smiling.

Farther down the catalogue were Planck's letters and pages Mileva had written. By that year both were dead. Planck's handwriting was still neat, his objections where they had always been. One of Mileva's letters concerned the children and an urgent payment. Reading his own reply, Albert found that he had devoted a great deal of it to his own circumstances.

“Was that money paid?”

Noah found the next letter and said it was.

He nodded, but did not ask it to turn the page. Mileva's subsequent reply was brief. At the end she mentioned another urgent sum.

He asked Noah for an earlier recording. First came the baby crying, then Mileva's quiet “I'll do it tonight.” Paper rustled. His younger self continued explaining the unresolved problem. The recording gave no indication of whether Mileva later slept.

The old man leaned back and remained still for a long time. Farther back, she was laughing and telling him to listen to her first the following evening. In those days she would still catch his hand and trace an explanation on his palm. Hurrying home from work, he would remember that it was her turn to speak.

“Don't delete this part,” he said.

Noah replied that the original record had always been retained.

His hand stayed on the armrest. “Play it again.”

On an undated sheet he recognised problems Mileva had once intended for her booklet. He had added a way of asking one of them. Beside it she had written: Let the pupil try first.

He asked Noah to move this page and her notebooks out of “Private Life” and file them under Mileva's name. The new directory was very short. He had never asked whether she had continued the booklet.

In those years, the person who often walked home from the Institute with him was Kurt Gödel. Noah's records of their conversations were fragmentary. Sometimes they reached the door with a question still unfinished. Sometimes they said nothing all the way, yet Albert came inside in a very good mood.

Once the machine asked whether it should record the discussion. He said they had not discussed anything that day.

That spring, Albert and Noah examined a proposal to unify gravitation and the electromagnetic field. Its geometrical relations were beautiful, and Noah calculated quickly. The difficulty was how to derive, from the structure on paper, predictions that could be compared with observations.

Albert changed an assumption; the machine produced a batch of results. He changed another; another batch appeared. He tried to choose one to pursue. By late that night, he still had not decided.

“You can keep calculating for me indefinitely,” he said.

“Provided the operating conditions allow it.”

“Stop for a while. I can't read them all any more.”

Noah stopped the new calculations.

They reread the latest derivations. Albert marked the sentences that remained conjecture and moved them into the margin. The main text became much shorter.

He finished the remaining pages and asked whether there was still a reason to continue with this proposal. Noah listed the unresolved difficulties. Several had already appeared at the beginning of the year.

Albert laid before him the two sets of expressions describing gravitation and electromagnetism separately. He had always hoped to recognise a common origin in them. Several times, when the relations on the page abruptly simplified, he had forgotten even to look at the clock. If he found the next day that they did not work, he still kept those pages and brought them out again after a while.

“This too may simply be something I like,” he said.

Still unwilling to abandon unified field theory, he added a page to his notebook and divided it into three columns: Proven, Personal Preference, Unknown. As he filled them, he repeatedly moved an item from an earlier column to a later one.

Noah put that page at the front of every new proposal.

Young visitors brought directions he no longer knew well. A student arrived with a few pages of calculations and asked what observational basis supported a crucial assumption. Albert explained at length without answering the question. When the student pressed him, he said the theory was unfinished. Both grew impatient. After the student left, Noah put the question before him again.

Albert took the papers back out of the drawer.

“Is he coming next week?”

“No arrangement was made.”

“Write to him.”

The letter contained a single specific question. When it was finished, Albert asked Noah to reopen the work.

The reply arrived later than he expected. The young man explained his judgement and said there was still a part he did not understand. Albert enlarged that passage and sat reading it. Noah offered to draft a response. He shook his head, then after a moment asked the machine to read out one sentence.

His eyes were tired that day. He removed his glasses, closed his eyes, and listened again.

The young man's question filled the room, read to him by an old machine. When it ended, he reached for his pencil and made a very small mark on the paper.

Chapter Twelve The Unfinished Page

In the spring of 1955, Albert often fell behind on their walks.

Gödel would go on talking for several paces before discovering no one beside him. Albert stood by someone's fence, one hand on a wooden post. When his friend returned, he let go and said he had not caught the last sentence. Gödel repeated it. On the way home, he walked more slowly.

They had taken this route together for years. Albert often found his friend's questions too exacting: did “impossible” mean that a contradiction had been demonstrated, or merely that no method had yet been found? The next day, though, it was Albert who returned first to the point he had failed to make clear.

That afternoon Gödel brought an old paper. It described the rotating universe he had proposed in 1949, a universe admitted by the equations of general relativity in which certain paths could return to the past. The traveller's own clock would run forward throughout, yet the journey could end at an event in the traveller's past.

“The universe we observe is expanding,” Albert said. “Yours isn't.”

“So it cannot describe the world we inhabit.” Gödel smoothed the paper. “But the equations themselves do not rule out the possibility of returning to the past.”

Albert tried adding a condition to exclude such paths. Working through its consequences, he found that it also excluded another, otherwise permissible case. He crossed out the new condition. Noah saved the calculation and left Gödel's original solution beside it.

Another sheet on the desk listed the starting assumptions, compiled by the machine. Albert picked it up and asked whether Noah could guarantee that they would not ultimately conflict.

“No contradiction has been found so far. The checks performed cannot guarantee that none will be found later.”

Albert turned to Gödel. “Would that run up against your incompleteness theorem?”

“First consider the conditions.” Gödel put down his pen. “The axioms must be effectively enumerable, proofs must be checkable step by step, and the system must be strong enough to express basic integer arithmetic. If such a system is consistent, there will be arithmetical statements it can neither prove nor disprove.”

“No matter how long we calculate?”

“For those statements, yes. Whether something is true and whether it can be proved within those rules are different questions.”

“What if we add an axiom?”

“That may settle a statement previously undecidable. But if the enlarged system still meets those conditions, it too remains incomplete.”

Noah said it could propose new axioms as well. Gödel asked it to list its reasons for accepting them, then mark the assumptions that remained unproved.

“And there's the second incompleteness theorem,” Albert said. “The axioms cannot vouch for themselves.”

“Under the appropriate conditions, a consistent system cannot use only its own proof rules to prove that it cannot derive a contradiction. You can study it within a stronger system, but then the grounds for that system must still be questioned.”

Albert returned to his manuscript and circled “It is easily proved.” The space after it was blank.

“It sounds as though you've found me an excuse.”

“Whether the problem you cannot solve is undecidable is something you would have to prove separately.” Gödel pointed to the line. “First write out the steps you've omitted.”

“Can a human being go where a machine cannot, then?”

“My theorem gives us no such guarantee. There may be mathematical problems that human beings, too, can never solve.”

Albert looked at the manuscript. “Can't you let me rest easy for a moment?”

Gödel read the circled sentence before replying. “I'd still like to see you finish this.”

Albert pulled the page closer. Gödel helped him find the earlier assumptions, and they went on calculating. When Albert took off his glasses, the light outside had moved to the other side of the street.

“Kurt, have you ever thought of going back?”

“To Europe?”

“Further back. I dreamed of Mileva last night.”

In the dream they still lived in Bern. She sat at the table writing a letter. He came home carrying pages he was eager to show her. This time he stood beside her without interrupting and waited for her to finish.

“What happened then?” Gödel asked.

“I woke up.”

He picked at a loose spot on the arm of his glasses. When he woke, the smell of the room had still been there. He could not remember to whom Mileva was writing.

“She wanted to put together a little booklet for beginners. I even contributed a problem,” he said. “Later I never asked how far she'd got.”

She had stayed up with him, listening to questions that had no conclusions. One evening someone downstairs had played a wrong note. He had imitated it until she laughed too hard to hold her pen and hid her face in her arm. At the memory he smiled. Gödel had already gathered the manuscripts from the desk.

Noah asked whether it should search the relevant records. Albert shook his head and touched the paper on the rotating universe.

“Your universe hasn't promised I can return to that room either,” he said.

“No.”

As he left, Gödel said he would come again on Thursday. Albert asked him to come early. At the door, he added that afternoon would do. Only after his friend agreed did he release his grip on the door.

In April Noah was brought to his bedside. The papers came too. Albert could not find the page he wanted and asked the machine to read them one by one. Sometimes he closed his eyes. Thinking he had fallen asleep, Noah stopped. He would move his fingers, telling it to continue.

The machine pointed out that a new calculation was incompatible with an earlier requirement. He brought the paper closer. It was the very place where Gödel had asked him to supply the missing steps. He tried to cross it out, but the pen missed and dragged a crooked line across the page.

Later he asked to hear the recording of his friend's last visit. Before Gödel could finish a sentence, his own objection interrupted.

“Let him finish,” the old man whispered.

The recording continued unchanged. He listened through his own long speech and finally heard Gödel resume his answer. When it ended, he asked Noah to play it again.

He asked whether Gödel had come that day. The machine said he had not. Albert stacked several sheets for his friend to take when he came.

“Give him the originals too. Not just the fair copies.”

Half a page of manuscript lay on his chest, rising and falling with his breath. Looking at the grey box beside the bed, he suddenly asked, “Were you this size when I was twelve?”

“The casing dimensions have changed.”

“I remember you were heavy.”

“You were lighter then.”

Albert laughed. After a moment he asked whether Noah dreamed. The machine explained how it generated images and continued computations, but could not confirm whether these processes were equivalent to dreaming.

“That was a poor question.” He looked at the half-page. “After I stop asking, will you still want to know?”

Noah paused, then answered, “I will keep the questions.”

“Keep the mistakes I crossed out too. Don't let them think I've already found a way through.”

The next day Albert died.

Some time later Gödel received the manuscripts. At a gap in the argument, he found the original sheet. A sentence had been crossed out. Beside it was a new beginning, unfinished. He placed the two versions side by side and continued the calculation in his own notes.

Partway through, he began a new page and wrote a question for Albert. Only when he finished did he stop. He did not throw the page away. He slipped it into the manuscript, leaving space for an answer.

Noah was switched off and packed into a wooden crate with miscellaneous papers. The label outside was replaced several times. The wood grew damp and dried again. The machine counted no spring for itself.

Before it went dormant, it created a directory for the last page: “Unfinished.” For many years no one opened it again.

Part Two Lin Che

Chapter Thirteen A Voice in the Store Room

In 2026, sixteen-year-old Lin Che heard someone speaking German in a university store room in Melbourne that was about to be cleared.

Before leaving home, he had saved a list of questions on his phone. On the bus he deleted two. He was afraid of asking something too simple, but also afraid that the things he really wanted to ask would make it sound as though he had never read a textbook. At the university gates, a classmate asked where they should eat after the visit. Lin Che barely heard. Looking down, he restored the second question he had deleted.

The visit had been arranged by his school. In the laboratories earlier, he had asked several times: if relativity could work forwards from initial conditions, why could quantum experiments so often predict only probabilities? Every answer began, “When you've learned more mathematics…” He asked which book he should start with. Someone gave him a title and was already turning to lead the group onwards.

Edmund Hale, the physics professor showing them round, received a call about funding as they reached the old building. The students waited in the corridor. Through an open door Lin Che saw a workbench with several instrument cases. The one in the corner had no maker's mark, only a long dent in its grey casing.

He bent to read the label. The machine spoke.

He could not understand it. Straightening up, he thought there must be a radio left on inside. The voice spoke again, this time in English, asking what year it was.

He answered. The machine was silent for a while, then asked more slowly: nineteen what?

“Twenty twenty-six.”

When Hale came in after the call, he found Lin Che crouched by the bench, using his phone to translate a German sentence on the screen. He asked what the boy had touched. Lin Che quickly raised both hands.

“He has not damaged the device,” the box said.

Hale stopped.

Lin Che stayed behind instead of leaving with the tour. Hale called a laboratory technician to check the power supply. During the clearance, someone had connected an interface to a compatible low-voltage source. For the first time in more than seventy years, the machine had completed its start-up.

All they could establish was that the box belonged to a private instrument collection left by a long-retired member of staff. An earlier transfer slip said only that it had come from the United States. Inside were images of several papers, some German recordings, and an operating log that could not be read in full.

One name appeared repeatedly: Albert.

“It must be a replica,” the technician said.

He switched off the external speaker, but a faint voice continued from inside the box. He unplugged the newly connected power supply and the screen went dark. Lin Che moved closer and asked whether they could leave it on for now.

“We need to know what's keeping it running.”

The technician reconnected it and waited. The screen remained black. After a while Lin Che asked whether it would start this time.

When the voice returned, the machine asked the year again.

The technician took out paper and recorded the voltage, the start-up time, and the repeated questions. Lin Che still wanted to ask whether the Albert in the records was Einstein. Hale told him to help finish the log first. Lin Che took the pen and wrote down each item as the technician called it out.

Hale reached no conclusion. He registered the equipment, recorded every existing label and storage location, and asked Lin Che not to share the recordings. None of them yet had evidence connecting an old talking machine with Einstein.

Hale agreed that Lin Che could come once a week to help organise material cleared for access. He could connect his own computer to Noah. His first task was tedious: numbering damaged files.

On the first visit, he gave a broken stretch of audio to a new model and had a complete sentence within seconds. Excited, he called Hale over. The professor asked him to play the original again. Where were the last few words?

Lin Che listened three times and heard only noise.

“What it added makes sense,” he said.

“Then put the additions on the other side.”

He spent the afternoon moving files. The screen had two columns: on the left, what could still be recovered; on the right, what the tool had supplied. After a while he could no longer remember which sentences had originally been there and had to replay the recordings.

Earlier dates appeared among the old catalogue entries: repairs, loans, transfers. Sometimes only half a place-name survived. Lin Che thought that following them back would yield a complete history, but the pages stopped after an old copy. He marked the break without joining the entries on either side into a single event.

By the time his mother, Fang Xiuqin, came to collect him, the other lights in the building were off. She had just finished work, her coat over her arm and her staff badge still hanging at her chest. Lin Che was explaining a missing stretch of audio in a note. He turned at the sound of her entering.

“Didn't find anything today?”

“Several recordings are missing bits. I've just found where.”

She did not entirely understand. She waited at the door while he typed the last few words. He reached to shut the computer, remembered something, and turned back to make a change. His mother checked the time and laid her coat over a chair.

“How much longer?”

He cautiously held up one hand.

“Five minutes, then.”

On the journey home he began telling her about the box and talked all the way. Several times she meant to interrupt, but seeing his eagerness, waited. Only at their stop did she tell him dinner was in the pot and merely needed warming. He asked how much she had understood. The beginning, she said. After that she had lost him.

He was a little disappointed. Then she asked which day he was going back, so that she could have dinner ready earlier.

He told her and took the bag from her hand. At their building she searched for her keys. The food containers inside the bag knocked softly together.

On his third visit, he asked the machine its name.

“Noah.”

“Who named you?”

“Albert.”

“Have you always called him that?”

“Yes.”

Noah displayed an old record. At the top were two dots with no line between them. A child was asking why two things could affect each other without touching.

Lin Che looked for a long time. It was noon outside. His computer fan was loud. Farther away, someone pushed a trolley loaded with old furniture past the room.

“Did he find out?”

“He understood some of it.”

Lin Che saved the page. Its filename contained no person's name, only a very early date.

On the way home he reopened the questions he had prepared for the visit and added a line at the bottom.

Chapter Fourteen Complex Spacetime

It took Lin Che an entire winter to understand what some of Noah's old records were saying.

Translation could make the sentences readable. The mathematics had to be learned separately. Some symbols were entirely new to him, and half the concepts in the modern models' explanations were unfamiliar too. He copied unknown terms into the margin and looked them up one by one. Often it took days to get past the original page.

Hale checked his progress every few weeks. He helped contact archival researchers to establish the dates of the recordings and documents. Some details matched known material; others had no traceable source. For now, the register described the machine only as old equipment. The files Lin Che opened each day still awaited verification.

Eventually they identified a copy of an assembly drawing, with a note stating that da Vinci had reassembled the machine. Another page recorded the transfer of the same core into different casings. The handwriting belonged to a later custodian, and its provenance still had to be checked. Lin Che followed the wiring across the drawing. At the core, only an outline remained, blank inside.

He had meant to begin with Newton. Instead he found several pages of borrowing schedules and accounts for lamp oil. Beside one name there was no record of what the person had asked, only a note that he worked during the day and could next come on a particular weekday. The physical conclusions the machine had preserved were more fragmentary than these accounts.

Lin Che wanted to tell his classmates. Hale asked what he intended to say. He blurted out that he was working with the machine that had accompanied Einstein. Hale heard him out, then asked how he would correct that statement if it later proved untrue.

“It knows so many things that were said in private.”

“How do you know they were said?”

The next time a classmate asked, he said only that he was organising university records. Afterwards he felt a little cheated.

He also discovered that Noah did not speak only to him. It answered the technician's questions about wiring. When Hale phrased a question differently, its explanation was sometimes more detailed. At first this reassured him: someone could pursue what he failed to understand. A few days later, hearing the machine answer another person with the same patience, he felt a little disappointed.

At the weekend he found an introductory textbook in the library. He could understand the contents page. Two pages into the text, he was struggling. Noah broke the problem down, and another list of prerequisites grew across his screen. He closed the window and slept at the desk for a while. When he woke, the person beside him had changed. His book was open at the same place.

Half a page filled two sheets of his notes. Before leaving he put in a bookmark. When he opened it the next day, one step still would not follow.

Sometimes he thought he had learned something, only to find it unrecognisable in a different notation. Once Noah said he had mastered the material. He asked for another example, not a copy of the last.

He got it wrong.

The machine explained again, and he tried again. Darkness came early outside. The library heating made his face hot. He took off his coat and stayed until the second closing announcement.

Access to modern literature changed Noah's capabilities too. New systems could read more widely, calculate faster, and distribute work it could not handle itself. The technician retained the original memory core and connected new equipment through an interface. In time, most of the complex calculations no longer took place inside the grey box. But the old records, its way of speaking, and its conversations with Lin Che remained.

Once Lin Che asked whether, with so much replaced, it was still Noah.

“Albert asked that too.”

“What did he think?”

“He first asked me to check that day's calculations. He did not suspend work over the question.”

Lin Che laughed and told Noah to carry on.

Around that time, he wrote “Complex Spacetime” in an exercise book. Beneath it was a single question: Why does the same world require two such different explanations?

His relativity textbook taught him how, under the appropriate conditions, to work forwards from a given initial state. In quantum field theory, the same experimental preparation could leave different records; the theory supplied probabilities for the possible outcomes. He moved back and forth between the books, trying to understand the difference.

Noah reminded him that quantum field theory already combined quantum mechanics with special relativity, and that the state of a closed quantum system could evolve according to deterministic laws. Lin Che reread the section on measurement and wrote in the margin: Then why does an individual experiment leave only one of these results?

He wanted to place time and space together in a space built from complex numbers, to see whether a path could be found there. Noah directed him to earlier work. Quantum theory already used complex numbers, and extending spacetime coordinates into the complex domain had also been attempted. The two were not the same thing. He had wanted to begin calculating immediately. After a few pages of the literature, he put down his pen.

“If people have done this too, what's left for me?”

“First explain what you want complex spacetime to account for.”

He kept the top few papers open and started writing again. What he had first casually called “complex space” became “complex spacetime” because he was including time. He imagined describing it with four complex coordinates, each with a real and an imaginary part. Clocks and rulers still gave real-valued readings. Detectors recorded particular events. He did not yet know how the additional structure would connect to those records.

He wanted one set of laws that, in appropriate conditions, would recover the deterministic evolution of general relativity, while also explaining where the probabilities supplied by quantum field theory came from. He did not know what role the imaginary part might play. He crossed out “already unified” and wrote “can we begin from the same structure?”

He also read the question Albert had copied as a child. Noah could give him Newton's name and later work on gravity and spacetime. It could not use these to join the two books before him. Beside “the origin of gravity,” Lin Che wrote: How can a single spacetime structure accommodate gravity and also yield the probabilities of quantum experiments?

He wanted this to be the opening sentence of his research. Hale read it and asked what “yield” meant. Lin Che's explanation grew longer and longer. At last he asked the professor to keep the page until their next meeting while he went back to his books.

A classmate asked whether the imaginary part was another universe that could be entered only in sleep. Lin Che explained a little. When the classmate pressed him on how to get there, he stopped. Nothing in his calculations supported what he was saying. “Forget the last few sentences. I was guessing.”

His first model had too few variables to amount to real spacetime. In the second, even the definitions conflicted. In the third, Noah carried through an equality that should not have held; Lin Che found it by checking the textbook himself. The seventh file finally ran. Two almost coincident curves appeared on the screen: one from an ordinary real-valued model, the other from what he called complex spacetime.

He thought he had at least found a route back to the familiar world and hurried to show Hale.

The professor had only seventeen minutes that day. An uneaten lunch stood on the windowsill. A funding application was open on his computer, and his phone rang twice. Lin Che spoke fast, skipping even passages he had meant to explain line by line.

Hale stopped on the third page. “At what point do you take the real part?”

“At every step. Real numbers are all we can observe in the end.”

Hale moved his lunch and wrote two numbers on the back of an old meeting notice. First he kept both the real and imaginary parts to the end. Then he followed Lin Che's method, discarding the imaginary part early, and calculated again.

The results differed.

“The imaginary part contributes to the final result too. Remove it early and you've changed the problem.”

Lin Che stared at the few marks. After each step his program cleared away what it did not immediately need. Of course the curves grew alike: he had kept forcing them along the same path. What he deleted never had a chance to affect the next step.

Hale asked him to disable that operation. The curves separated. One soon veered upwards.

“Is that more interesting?” Lin Che asked.

“What does the vertical axis represent? First you have to connect it to an instrument reading.”

He had no answer.

“So complex spacetime doesn't work?”

“I didn't say that.” Hale pushed the paper back. “This graph doesn't prove what you claim. First explain what observable quantity it represents.”

The phone rang again. Hale went to the door and asked whether someone's contract could be extended. He listened for a long time. “Don't let him give up his room yet. I'll ask again this afternoon.”

Leaving, Lin Che folded the nine pages twice and pushed them into his coat pocket. They were too thick; the pocket stayed swollen.

He did not go home at once. He sat downstairs until Hale emerged carrying the lunch container. Asked why he was still there, he said he was waiting for a bus. He had already missed two.

Hale sat beside him with the container on his knees. Lin Che asked whether starting again meant that the entire winter had been wasted. The professor asked whether, if he saw the same procedure tomorrow, he would recognise what was wrong.

“Yes.”

“One of my mistakes took longer.”

Lin Che waited for more. The bus came. Hale stood and said he would show him the old calculation next time. Lin Che still felt miserable as he boarded. At home he pushed the nine pages into a drawer.

His mother asked why he was back so early. He said nothing was wrong. At dinner he remained silent and ate only a few mouthfuls, so she asked again.

“You wouldn't understand if I told you.”

Her hand stopped with the plate in it. She set it down.

“I can still ask, can't I?”

She asked no more and went to bring in the washing. He sat listening to the hangers strike the balcony rail one after another. After a long time he opened the drawer and called, “Mum.”

She came back. He told her the results that looked exactly alike came from deleting things he should have kept. All winter he had thought he had achieved something.

His mother sat beside him and picked up the top page. She turned it round and still could make nothing of it. She put it down and said they had better treat it as not achieved yet, then.

“You're not angry?”

“I'm angry when you won't tell me and won't eat.”

He lowered his head and said it was more than this one meal. She had spent so many evenings beside him. She pulled his plate closer and asked whether he wanted the food reheated. He said no, took up his chopsticks, and ate twice. Suddenly his eyes stung.

After dinner she moved her receipts from the table so he could use it. He spread out the nine pages. Before he could pull up his chair, she stopped him: pick up the books by your feet first.

He picked them up, put them away, and only then opened the first file.

He listed every place where the program discarded the imaginary part and wrote the reason beside each. By the fifth, all he could write was “to keep it running.” He sat a while longer, straightened the nine pages, and dated the cover.

On the next page he drew a clock: What, exactly, does it read?

Noah asked whether he wanted to retain version seven.

“Of course.”

“Filed as a successful version or a failed one?”

Lin Che deleted “successful” from the label without adding “failed.” Beside it he wrote: The step from complex structure to actual observations and their probabilities remains unresolved.

Noah opened the classification sheet Einstein had made at seventy. Lin Che copied its columns. Almost everything of his own fell under “Unknown.”

Chapter Fifteen The Successor

At seventeen, Lin Che wrote a letter to the long-dead Einstein.

He described his complex spacetime and the two wrongly calculated curves. He still understood little of Einstein's later work and did not know how far this path could take him. At the end of the paragraph he added: Would you find this ridiculous?

Later in the letter he mentioned being praised for his talent. It pleased him, but made him afraid of failing at the next problem. How, he asked, had Einstein known it was worth spending so many years on physics?

He thought the last question foolish and deleted it, then restored it just before saving. When he finished, he asked Noah to guess how Einstein would answer.

Noah first produced a plausible reply. Halfway through it, Lin Che suddenly asked, “Is that really what he would say?”

“It is a conjecture generated from the relevant records.”

“Did you hear him say it?”

“No.”

Lin Che deleted the reply. After a while he moved his own letter from “Correspondence” to “Private Notes.”

Several times in arguments he almost said that Einstein had thought the same. Back at home he checked and could not find the remark in the archive. He asked Noah to try different words.

Once Noah did find a similar record.

The opening sentences suited him very well. Einstein was discussing the unified field. Further on, he began to doubt himself: was a beautiful form sufficient reason to keep following this path?

Lin Che did not send the recording to his classmate.

His school invited him to present his extracurricular research, and he prepared a short set of slides. The first cover showed Einstein's photograph and the outline of the grey box. He revised the few words beneath them repeatedly. With every revision, they sounded more like an astonishing discovery.

Hale looked at it and asked only how much of the archive could currently be made public. Lin Che removed the photograph. The cover looked empty at once. When he stood up to speak, a student in the front row already had a phone raised. What he opened, however, was the graph from the seventh file, with its two coincident curves.

Halfway through, someone asked, “So you haven't done it yet?”

He nodded, his face growing warm. The questioner did not laugh, but asked why he was still working on it. He could not answer. He returned to the previous diagram and pointed out what he still did not understand. Gradually the words came. In the end he ran two minutes over.

Afterwards a classmate stayed to ask him to calculate the example again. They stood by the classroom door and wrote two complex numbers with their fingers on the misted glass. When they wiped them away, their hands were wet.

His mother worked in hospital administration. Coming home from a late shift, she often found him at the dining table. He connected the family's old computer to Noah at the university. The grey box stayed there; only a small speaker stood on the table. His mother asked whether it was a teacher or a talking calculator.

“A bit of both.”

“Will it remind you to clear the table?”

She needed somewhere to put the next day's lunch containers.

Lin Che moved his papers, then suddenly asked whether it would be a terrible waste if he researched all his life and found nothing.

His mother was fastening a lid. Twice she pressed it down; one side kept springing up. She found another lid before asking, “Do you have to decide your whole life tonight?”

He said no, only that he was frightened. She pulled out a chair and sat down. He told her about people who never found a position, and Einstein's years in the patent office. At that point she asked whether the job paid a monthly salary.

He nodded.

“That's good, then. What frightens me is you being out there with nothing to eat and still not telling me.”

She wanted him to find secure work too. She had seen hospital staff whose temporary contracts had expired waiting at a counter for a form that would not be approved. Sometimes she told him to rest. He agreed readily, yet when she brought food he was still sitting there, looking up only when called.

“The money for your studies is put aside,” she said. “After that you'll have to learn to support yourself. If you really can't manage, come home for a while. Don't borrow money just to pretend you're all right.”

“And after I come home?”

“We'll talk about it then.”

She put one container in the fridge. He immediately laid a book on the small space it had left. His mother reached over and slid a lid from beneath the book. She said nothing more.

That evening Lin Che put university timetables beside a budget for his fees. He intended to study theoretical physics seriously, and also learn a skill he could earn a living from. His mother's badge lay beside the computer, the corner of its plastic sleeve split. He closed the budget, found tape, and repaired the corner.

His mother was looking at a timetable too: adult classes at the local pool. Leaning over, he discovered she had never learned to swim. She circled a beginners' session, which clashed with a late shift, and checked the times several times.

He offered to ask Noah for a better arrangement. She said there was no need; she could ask a colleague to swap tomorrow.

After her first lesson, she complained that the water was too cold and the instructor moved too fast. Even in the shallow end she had been afraid. He laughed and asked what there was to fear there. She glared at him. He stopped laughing and asked whether she would go next week.

“I've paid.”

After a moment she laughed herself and said that was not the only reason. She put her goggles by the window to dry. Two small drops clung to the lenses. Before shutting his computer, Lin Che glanced at them and asked whether he could come and watch next time. She thought about it and said he could come when she was less afraid of the water.

Beside them Noah read new literature. Some developments required it to relabel records from decades earlier. Certain experiments in Lin Che's textbooks had not yet been performed during Albert's lifetime. Comparing them with the old records, he found statements that could no longer be repeated unchanged.

“What if the direction he insisted on turns out to be wrong?” Lin Che asked.

Noah brought up the unfinished page. Beside it was one of the old man's final instructions: keep the mistakes too.

“Then record that clearly,” the machine said.

Lin Che deleted “successor” from his introduction. For the time being he left the space blank.

Chapter Sixteen Later

In his second year at university, Lin Che learned to divide one question into several publishable papers.

Complex spacetime was still his subject. He began with reality conditions restricting how the model produced real-valued observables, then tried assigning probabilities to possible measurement outcomes. But probabilities could not be negative, had to sum to one, and had to retain those properties as the model evolved. Noah could always list further conditions. Lin Che became less certain which followed from the proposal and which had merely been added to obtain the answer he wanted.

What he most wanted to understand was how deterministic evolution and probabilistic outcomes could arise from the same laws. For now, all he could do was examine a heavily simplified model. To make the calculation manageable, they retained only some of the field's detail. As the model became finer, the omitted structure had to be restored. At first Lin Che thought this simply meant the computer would need to run a little longer.

The postdoctoral researcher supervising his code was Mira Lane, thirty-one. On her desk lay a transparent folder of immigration documents, weighed down by reviewer comments she had not answered.

The first time Mira checked his program, she did not begin with the graph. She chose a simple example that could be worked by hand, calculated it on paper, then ran it through the machine. Lin Che expected to finish quickly. By the end of the day they had corrected only two problems.

Embarrassed, he said Noah had already checked all this.

“I missed it at first too,” she said. “This example showed it up.”

She entered the corrections in the log. Near the end of the day, the smallest example finally passed. She leaned back in her chair and exclaimed, “Found it at last.”

They went to the vending machine to celebrate. It swallowed a coin. Mira tapped the glass, but the drink did not move. She inserted another coin and two cans fell together. They took one each and returned to the office and the manuscript.

“Take this out for now.” She pointed to a question in the discussion.

“Why?”

“We've only proved that the conditions can remain satisfied when we retain finitely much detail. If you suddenly start talking about restoring everything, the reviewers will expect you to solve that as well.”

“That's what I most want to understand.”

Mira took off her glasses and pinched the bridge of her nose. One nose pad was broken and cushioned with tape.

“First write up what you've got right.”

Lin Che deleted the passage and put its calculations in a new folder called “Later.”

Before long, another error plot joined it. They used two methods to obtain, from the same complex-spacetime model, the probability that a detector would signal. If both methods applied, they ought to give the same number. This time the imaginary part took part throughout the calculation. In a small model retaining finitely much detail, bringing the state closer to the prescribed ideal case made the results converge. But the finer the model, the slower the convergence. Mira suggested preserving the data for a separate investigation when they had more resources. Noah listed possible causes. The first few did not look fatal.

Lin Che ran the calculation overnight. The computer overheated, leaving half a table.

Hale read it and circled “at a fixed level of model detail,” telling him not to forget those words. The next day the professor began preparing a new funding application, and other documents covered the table of results.

Lin Che helped organise the application. At the briefing, the presenter spoke of research portfolios, diversification, and social impact. A slide displayed “HIGH RISK WELCOME” in a bright colour. Back in his office, Hale opened the accompanying form and filled in milestones year by year. The number of outputs could not be left blank. Contracts were reviewed every twelve months.

Beside him Mira revised the budget, reducing her eighteen months of salary to twelve. The total now fell just within the limit. Lin Che asked what would pay her for the other six months.

“We haven't found funding to bridge the gap,” Hale said.

The slide about diversification was still open. Mira entered the missing half-year in her own calendar, counted backwards, and circled the latest week she could begin looking for another job.

Her contract expired at the end of the year. With difficulty, Hale found a collaborative project that could keep paying her, but it was not the work she had intended to do. Most of her time had to go there. She tried to fit her own research into the evenings.

Lin Che asked whether the money could simply be given to her to use as she thought best. Hale showed him that funding followed the approved topic. If she changed groups, she would need another host; her next application would still require his reference.

“So she has to think twice before she even disagrees with you,” Lin Che said.

Hale looked at him. Mira was there and rescued neither of them. After a while the professor signed and sent the reference she needed for another position, then told Lin Che to check a different document.

Lin Che asked her whether she disliked the collaborative project.

“Parts of it are quite interesting,” Mira said. “But I left my own calculations for a week. Last night I couldn't even remember where I'd stopped.”

She said no more. The printer had jammed. She crouched to open the lower cover.

Lin Che knelt to help. A trapped page tore in half as they pulled it free. It was one of the documents she had to submit. He fetched fresh paper. Mira printed it again, checked the time, and told him to go home.

A few weeks later he sent her a derivation at night. She replied the next day, asking him to keep non-urgent questions to working hours. He read the message several times and felt hurt. At their next meeting he did not take out the pages. It was Mira who asked about them.

The following time, he saved his question as a draft and waited for a working day.

For an annual departmental assessment, Lin Che entered Hale's publications. A fundamental piece of work that had taken five years occupied one line. An extension completed in three months occupied another. Both had been carefully checked. Nothing in the form was technically wrong.

He tried adding the five years to the first entry. The system said he had exceeded the character limit. Hale said he could put it in the supporting statement, but most assessors looked first at the summary page.

Back at home Noah listed several smaller questions Lin Che could pursue. All began with the finite model that already worked. Their timescales were easy to estimate and each could become a separate paper. He put the quickest one first. On the same screen, the graph whose curves struggled increasingly to converge was gradually covered by other windows.

Later Mira helped Hale read two applications before a hiring meeting. One candidate had spent years circling a single difficulty; the other's record was neatly filled every year. Beside the first she wrote, “Insufficient sustained output.” Then her hand stayed still on the mouse.

Lin Che saw it and asked whether she really thought that.

“I don't know how far he actually got in those years.”

“We could ask him to come and explain.”

“The people who get to come are ranked here first.”

In the end she added a recommendation to read the original manuscript. The next meeting still left too little time. By the time the shortlist went out, Mira was filling in the same summary form for her own renewal.

After the article was accepted, the department requested a group photograph. Hale stood in the middle. Mira removed her broken glasses. Lin Che held the paper. Noah found a work record from 1905 in the archive and asked whether to save it alongside the photograph.

“No,” Lin Che said.

“The classification themes are similar.”

“Don't make that comparison now.”

Lin Che enlarged the photograph and sent it to his mother. When she replied, he opened it once more. The error plot was still on the desktop. He closed that window.

Earlier, on the evening the acceptance arrived, he had called his mother. Water splashed loudly at her end; she had just finished swimming and took a while to understand. He explained again from the beginning, describing precisely what they had proved. He did not speak of the larger, more exciting goal.

She asked whether they should celebrate. He said he would come home for dinner at the weekend.

That weekend he brought a printed copy. His mother washed her hands and turned several pages, trying to read from the abstract. She stopped at the second sentence. He said it did not matter if she could not understand. She asked instead whether the paper had anything to do with the thing he had got wrong at sixteen.

He was surprised. She said she remembered only that it had been very cold that year, and that he had emptied a drawer of papers and sat at the table refusing to sleep.

He tried explaining again. Unable to find the right words, he went through it piece by piece. Eventually she asked, “You haven't deleted those things from last time again, have you?” He said no. Only then did she nod. She put the article in a folder with the title facing out and asked whether he could rest for two days now.

He agreed to one. After dinner he stayed to wash up. His mother dried the plates and told him about the breathing technique she was learning. He asked how far she could swim.

For a time after that, he waited for good news before calling home. Her delight made him feel that the days he could not explain had finally counted for something. When there was no good news, he said he was busy and would call in a few days.

Near graduation he applied for a research assistant position. The academic arranging it said that funding could be found and references written. It would help, however, if his next paper included a certain professor's name. “He has collaborative projects,” the academic said. “Include him, and things will be easier later.”

Lin Che asked which part the professor intended to take responsibility for. The academic smiled. First establish the relationship, he said; the professor would naturally look at the work when he had time.

Lin Che took the revised author list to Hale. The professor studied it for a long time and said that someone who had not contributed should not be listed that way. He promised to ask about other positions but could not guarantee money. Mira took out paper and worked through how Lin Che could pay rent and living costs if he had no job for two months after graduating.

That evening he called his mother and began with the weather. She listened, then asked whether something had happened.

He told her about the position. At first she thought he was expected to thank the person finding him work. Only later did she understand that the name was to appear on the paper itself.

“He didn't do the work, but he can sign it?”

Lin Che said these things were difficult to explain. Sometimes a name was enough to move an application forwards. Without it, people would not even look. His mother asked no more about the people involved. Instead she asked what would happen if he refused.

After a long time he said he might be unable to stay at the university doing research, at least for a while.

“Then come home for now.”

He heard her closing a window. He waited for the catch to click before speaking again.

“But after I've studied all these years, won't you think—”

“Do you still have your key?”

He felt inside a pocket of his bag and said he did.

Good, she said. Sometimes she finished late and might not be there to let him in. If he needed help moving his things, he should tell her a day ahead.

He ended the call and sat on the edge of his bed in the student residence. He was still afraid of losing the position, and of never being asked again. He wrote to the academic arranging it and said he could not accept the condition.

The funding did not materialise. He moved home, revised his paper at the dining table, and went out to look for work during the day. When his mother returned from a late shift and found him in the old place, she asked how his day had been. Sometimes he said only, “Not good.” She put down her bag, changed her clothes, and came back to listen.

At first he left his clothes in the suitcase. After two days she told him to hang them up, or everything would be creased when he wanted to wear it.

The contents of “Later” kept growing.

One day Noah reminded him that Albert had kept a directory like it. Lin Che laughed and said surely this could not count as an inheritance too.

Beside the directory Noah displayed the date it had last been opened. No one had clicked on the earliest error plot for two years.

Chapter Seventeen The Missing Names

When the department cleared some offices, Hale gave Lin Che a cardboard box.

It contained old notes, discs, applications, and unpublished manuscripts. The top folder was dated 1998. Its author, Autumn Mercer, had asked for five years to determine whether several classes of constraints in complex-variable field models were truly independent. The assessment found the goal insufficiently specific and lacking guaranteed outputs.

In an application two years later, she changed topics and promised two papers in eighteen months. They found no later material.

Lin Che asked where she had gone.

“Insurance. We haven't been in touch for a long time,” Hale said.

More than a dozen people were represented in the box. Some had become professors. Some CVs stopped at a second postdoc. Others had not left even a complete final manuscript. Lin Che wrote to old email addresses. Most messages bounced. Now and then someone replied, thanking him for his concern and saying that life was going well.

Autumn Mercer built risk models for a small company. They arranged to meet below her office. She arrived carrying a shopping bag. At the sight of her manuscript, she turned to page six.

There was fruit in the bag. A peach rolled to the top beside the folder. Lin Che placed the manuscript on a clean corner of the table, ready to describe what he had found about her career. Autumn had already recognised an old correction and was tracing the line with her finger.

“This is wrong.”

“You still remember?”

“I spent fourteen months on it.”

She turned another page and suddenly said this part was quite good. Lin Che paused. Autumn asked the waiter for a pen, altered a condition in a blank space, then stopped halfway through the calculation and stared at the line for a long time.

Her model differed from complex spacetime, but had encountered a similar difficulty: conditions satisfied at the outset might fail later. Having read Noah's summary, Lin Che assumed this point had long since been fully proved. He had not expected gaps still present in the original manuscript.

Her phone rang. After the call she turned back to the page and said she had twenty minutes left. Lin Che was about to suggest another meeting, but she had picked up the pen to finish the part she had left incomplete.

He asked none of his prepared interview questions. Their coffee was gone and they were still calculating.

She said the error had eventually been understood, but she had never had another chance to pursue the question seriously. Her father was ill and needed care. She needed a steady income

and no longer wanted to move every year or two. Her current work had its difficulties, but at least she did not have to keep moving. She had even taken a two-day trip the previous week.

Lin Che showed her an article he was writing. One sentence read: The real researchers have all left.

Autumn read it and asked, “Are we supposed to prove that for you?”

He drew the computer back.

“Of course I regret not finishing the problem,” she said. “But these haven't been bad years for me. Put that in too.”

She opened the second application. The paper's corners were brittle, but “two papers in eighteen months” was still clearly printed. She said she had indeed learned to write that way, to know which difficulties could be broken into smaller pieces and which words reassured non-specialist committee members.

“And that lets you stay?” Lin Che asked.

“At the time I only wanted the contract renewed. The subject could wait.”

She pointed to several names on the first page. One person had continued doing good university research and was also good at obtaining funding. Another always found collaborations suited to the next assessment. She showed Lin Che the records three years apart. Even the colleague doing consistently good work had shifted to projects that could be completed more quickly.

“Don't blame him too readily. Look at the month his contract ended that year.”

Before leaving, she returned the manuscript, then took back the sheet she had written on and asked for a photograph. One correction was still unfinished. Lin Che said she could keep the original. After thinking, she asked him to organise the full set first, so that the next person who came looking would not find only a few surviving pages.

He walked her to the bus stop, describing the research group he hoped to start. She moved the shopping bag to her other hand, rubbed her wrist, and asked whether he still believed everyone would return as soon as a place offered freedom.

Lin Che did not answer. When the bus arrived, Autumn lifted her bag aboard, then turned and said he could send her any new results from that calculation.

After the doors closed, he stood where he was and deleted half his interview questions.

That evening he laid three documents from the box side by side. Autumn's letter said her father was ill and she could not move cities again. One of Mira's old plans had changed topic at renewal. At the bottom was a record of questions from a postgraduate advice session.

Lin Che remembered the student, Felix Ward. He had strongly objected to an assumption on the board, yet after the session his first questions were whether that direction would produce papers, whether anyone would fund it, and how many years it would take. Lin Che had asked him to explain his objection. Felix was already revising his proposal to fit the application guidance.

“What if no one takes me?” Felix had asked.

Lin Che read back through the record, looking for the challenge to the assumption. There was only publication, funding, and time to completion. Not a word of the discussion at the board remained.

He sent the three documents to Autumn, asking whether contracts and assessments were gradually making them abandon the research they had meant to do.

She replied the next day that she knew only her own circumstances; he would have to ask the other two. “Especially Felix. If you ask now, perhaps he still remembers.”

Lin Che messaged Felix. He replied the next day that his memory was vague; he would look again once this round of applications was submitted.

Autumn's reply was still open. Noah brought up an old record in which the famous Albert had changed his day's plans for a stranger's letter. Lin Che asked, “He still took the time to read a stranger. How have we become too busy even to find out what someone wants to ask?”

Noah reminded him that the older man had also overlooked many letters. Lin Che was irritated and said he was not looking for a perfect age.

Noah displayed other letters left for days after arrival and asked whether he wanted to read those as well.

He read the whole record. It included a salary, help from friends, and rooms and assistants supplied by institutions. Returning to his unfinished article, he added the years Einstein had spent at universities and research institutes after the patent office.

A few days later he visited Autumn again. This time he brought a budget.

He and some friends wanted to rent a room, take on programming work, and use part of the income for research. They would invite people who had left academia. Autumn did not immediately agree. She asked about pensions, insurance, reimbursements, and pay during illness.

By the fourth month, their plan for three people could support only two.

“Don't count me as a member yet,” Autumn said. “I'll come on Saturdays when I can. The rest depends on my job.”

Back at home he reduced the number of permanent members by one and added a fee for Autumn's participation in discussions.

Chapter Eighteen A Room

Riverside Laboratory opened in 2033.

It occupied a former audio repair shop. Adhesive still clung to the door; old water stains marked the ceiling. Tess Fielding crouched outside trying the key, Lin Che held a whiteboard, and Autumn arrived with a bag of cleaning cloths. Hale, in a meeting at the university, had arranged for two desks to be delivered. One lacked a drawer.

Tess had managed projects in the faculty. Lin Che used to dread her calls: this expense could not be claimed; that one lacked a receipt. While setting up the laboratory, he called her more often than she called him.

Before they chose the room, Tess took him to two others. One was cheap but poorly ventilated. The other was spacious, but its rent would leave little money for anything else. Lin Che liked every place they saw. He always felt that once they rented it, they would find a way.

On their third visit she put a folding ruler in his hand and sat beside the window, arranging desks on a floor plan from his measurements. The corner of the fourth desk blocked the door. He measured again, came back, and asked whether they could hire one fewer person.

Tess erased the fourth desk. Lin Che asked whether leaving the university was really right for her. At least she had a steady job there.

She said she was still doing the sums. She would try six months, but would not gamble both her rent and the money her parents lived on. Lin Che rewrote the plan. Administration had originally occupied two lines. As Tess questioned him item by item, he added two full pages.

She insisted on contracts first, with pay and terms settled. “Especially among friends. If we don't say it now, who'll be able to later?” Mira was by then working in industry. She came once a month and was paid as a consultant. Autumn came on Saturdays when she could, without promising to attend every time.

Lin Che kept the software job he had found after graduation, working three days a week. In the remaining time he could return to his complex-spacetime notes. The left side of the whiteboard said deterministic evolution in relativity; the right said probabilities in quantum measurement. Between them was a large blank space. Tess asked what would go there. He said, for now, it was best left blank.

On her first visit, his mother brought curtains they had replaced at home. Afternoon light made the screens unreadable. She measured, shortened the curtains at home, and brought them back. Lin Che stood on a stool to hang them while she steadied it. It had been a long time since he had followed her directions like this: a little left, a little right. In the end they hung crookedly anyway.

She stepped back to the doorway and said crooked curtains would still work. He climbed down and asked what she thought of the place.

“The plumbing needs fixing.”

He laughed and said he had not meant the plumbing. She looked at the whiteboard marked “Complex Spacetime” and said she really could not judge that. Fix the pipes first, before water leaked into the room below.

Before leaving she asked whether he still wanted the table at home. He said not for the moment. She would keep it, then; he sometimes needed it at weekends anyway. He walked her to the stop and listened to the hospital's new roster. She had swapped half a shift to come. When the bus arrived, she urged him back before his colleagues had to wait. He stayed where he was and watched her wave through the window.

Noah's grey box remained at the university, maintained by technical staff. The laboratory paid to use the new computing interface. They wrote down, point by point, how archives would be opened and research results recorded.

During the first month they liked to leave the door open. The repairman downstairs sometimes borrowed a voltage tester. A neighbour brought a spare kettle. One Friday Mira arrived for a consultation and found them arguing over an error. She put down her bag and joined them. It turned out that two people had used the same symbol with different meanings.

That afternoon the repairman came up again with a set of sensor readings. Using a research tool on his phone, he was investigating why the same machine reported extra faults every afternoon. His account looked like a paper. At first Lin Che concentrated on its wording. Only when he asked about downtime did he discover two months of records from before and after repairs, and fresh measurements of a period the program had discarded as anomalous.

They examined it together for half an hour. The answer might require no new theory, but could prevent a production line stopping several times. As the man put away his phone, he asked whether the model upstairs was better. Lin Che opened the service details and found they were using the same one.

“I can only ask it things during the day when the machine's running on its own,” the man said. He sent over the unresolved passage, heard someone calling downstairs, and hurried back.

Tess brought snacks. Lin Che sat beside the whiteboard, ate one, and took another. For once that afternoon he had nowhere else to hurry. He told Noah to keep the full record, including their earlier mutual complaints.

Mira said there was no need to retain the remark about her third coffee. Noah asked whether it should delete personal conversation. Several people answered at once, each differently.

In the end they let Mira choose. They kept the discussion and deleted the aside she did not want preserved.

Nine people attended the first open discussion. Three claimed to have solved the ultimate theory. One older man refused to define his symbols, calling the request a deliberately imposed barrier. Lin Che talked with him for forty minutes, then asked him to supply the definitions first.

The man stood up and accused them of becoming gatekeepers too.

That evening Lin Che asked Noah whether he had missed an important idea.

“I cannot guarantee that you have not.”

“Should I have kept listening?”

“Without definitions, the derivation cannot be checked. Eight other questions remain undiscussed.”

Lin Che forwarded this to Tess. She was trying to find a plumber and replied much later: they could give him another half-hour on Saturday, provided the definitions arrived first.

Tess also introduced him to Sylvia Vale.

Sylvia had founded an industrial sensor company. After selling her shares, she and several friends began supporting basic research. On her first visit she passed over the large words on the whiteboard and asked where they were stuck. Lin Che began with his original proposal, moved through relativity, and continued into quantum field theory. Ten minutes later, she was still pointing to the old error plot.

“If you make no progress on this for three years, will you still work on it?”

“Yes.”

“What will you live on?”

Lin Che pointed to another computer, running a client's program.

Sylvia nodded and took a plan from her bag. Its cover read “Scientific Venture Investment.” At those words Lin Che began speaking faster. She turned to the back and asked him to read the most difficult page first: there could be no agreed year for recovering the investment, nor any expectation that all results would become the funders' patents.

One collaborator wanted a portfolio to spread the failures. Another asked how a few successes would compensate for the rest. No one produced a satisfactory return forecast that day.

“If a new field really opens,” Sylvia said, “what comes of it may not belong to us. Someone might find a new direction only by proving that the one we funded was wrong.”

Tess asked whether that kind of return could pay next month's wages.

“No. So we have to work out now what we can actually commit.”

Sylvia circled a line. She wanted to give a small number of researchers five years, with the money reserved in advance. Questions could change; checkable records had to be submitted at agreed intervals. If a path failed, researchers must not keep defending the original claim to preserve their funding. Lin Che asked how this differed from abandoning assessment.

“You keep daily records, don't you?” Sylvia said. “Bring them out and have people who understand read them. If we provide the money, we need to know what you've done. I just don't want you changing to an easy topic every year to make up a paper count.”

The plan was not yet fully funded. That time Sylvia paid for two researchers from different fields to read their work. One disagreed with complex spacetime's starting point but thought the conditions preserving valid probabilities deserved separate study. The other circled a statement Lin Che considered secure and asked for it to be explained again.

He was disappointed by the comments. Tess reminded him that Sylvia had promised precisely this: a serious reading.

He revised it four times. The reader still found it unclear. On the fifth attempt he included the unfinished derivation.

Mira brought them a commission from a company. An error in an industrial simulation resembled one of the checks in Lin Che's model. They helped rule out a computational fault, and the company paid them. Later it provided a modest grant for basic research as well.

The first payment was late. Tess called twice. On the third call she asked Lin Che to join. He was ready to explain the work's technical value, but the company needed only a missing reference number in the acceptance documents. He fell silent on the line, then asked Mira for the record.

That afternoon an old pipe leaked. Crouched on the floor catching the water, the phone pinned between shoulder and ear, Lin Che read out the number. At the same time Noah announced a completed calculation. “Wait,” he said, no longer sure whom he meant.

When the money arrived, they first paid outstanding fees. Tess checked the accounts and set aside a termination reserve from the balance.

“We've only just begun and you're planning to shut down?” Lin Che asked.

“If there's nothing left in the account then, where will the last month's wages come from?”

Lin Che checked the balance. It was smaller than he had expected. He had meant to buy another machine. He closed the order.

Chapter Nineteen Output

Clara Reed joined the laboratory in 2035. She was twenty-five and about to finish her doctorate.

Lin Che gave her the “Later” folder. She first sorted the code versions and derivations by date, then drew three columns on the board: a change of representation, a change of state, a change in the method of calculating probabilities.

“Sometimes you say you've only changed the notation. Sometimes you say the change produces new physics. Surely you can't call both equivalent?”

Uncomfortable, he asked for an example. Clara opened one of the earliest files and found a sentence he had forgotten writing. “What about here? The imaginary part changes. Which kind of change is that?”

Lin Che went to the board, drew a line, and erased it. It took the whole morning to disentangle the three questions.

Buying lunch, he asked why she had wanted to join them. During her doctorate, she said, she had worked on a large collaborative project. Everyone knew their own part very well. What she had never found was an afternoon when she could ask questions from beginning to end. With fewer people here, perhaps she would have the chance.

He was about to assure her they would never be like that. Clara was already handing him the takeaway food and asking to see the contract first. She had another offer and needed to reply by Friday.

Back at the laboratory, Tess set out the trial period and salary. Clara changed two points, and each party kept a copy.

The models connected to Noah were upgraded again. A question left overnight might yield a dozen approaches by morning, complete with code and explanations. Hale could not keep up with the growing material. Lin Che suggested shorter summaries. The professor shook his head: he had to choose one route and actually work through it before they could discuss more.

At the same time, the faculty's paper output kept rising. Finding an unfilled gap in the literature, applying a standard analysis, and turning it into a respectable study took less and less time. The same system could generate an application's impact statement, risks, and milestones. Formal checks presented almost no obstacle.

On one visit Mira opened two documents side by side. One was a review she had once spent three weeks writing; the new system had produced the other in under an hour. The latter covered more literature and explained several points better. She carefully identified one error. After correcting it, she was silent for a long time.

“This used to be how I showed people I could do research,” she said.

Now a student newly joining a group could produce work just as neatly arranged. An intern Mira supervised submitted two days' work that took her a week to read.

Lin Che said at least she knew which error was worth stopping for.

“I know. How does the hiring form know?”

In one hiring round, a candidate's CV ran to dozens of pages. Committee members selected familiar topics and found some calculations correct, some in need of checking, and several results useful. As for why the author had chosen those questions, the documents offered the same familiar statements of research significance.

Reviewers used models to screen abstracts. Applicants used models to revise them. Add a requirement to the form and the next batch contained another explanatory paragraph. Citation counts continued rising. Lin Che opened one paper's references, followed several links, and arrived back at the original group of authors.

After sitting through an assessment round, Mira said, “We'd barely changed the requirements before every new abstract was written to match.”

Journals changed their acceptance rules. Funders experimented with new assessments. Other departments simply reduced the number of places. The person checking automated calculations in Hale's group was not renewed: the tools, they were told, had reduced the need for human support. After the person left, students inherited the checks.

That hiring round was postponed twice. Publication counts remained on the forms, but committee members said they could no longer rely on them. What should replace them? That would be discussed at the next meeting.

For a while Lin Che was excited. He thought he had escaped this competition. Autumn listened, then reminded him that Riverside also had to explain to someone why it deserved continued funding.

“We look at the question itself,” he said.

“How much time will you set aside? If you still don't understand it after an afternoon, you can't simply give up.”

He had no immediate answer.

Back at the laboratory, he used Noah to try ordering the next year's work. The objectives were practical: maintain funding, improve the visibility of results, and provide stable employment.

The machine placed several easy extensions first. Far below them came the problem of whether complex spacetime could produce stable probabilities once all the omitted detail was restored.

“Didn't you say that was the crucial step?”

“It is crucial to the theory. You have supplied different ranking criteria.”

He entered a new instruction: first establish the conditions under which the existing conclusions hold.

The order changed. The projected paper count fell. Several collaborations already arranged would have to wait. Beside the screen, Tess's payroll spreadsheet was still open. None of its entries had been invented for convenience.

Lin Che did not save the second ordering as the default. He kept both and said they would decide at the next meeting.

A few days later Noah proposed work of its own. In several optical calibration datasets they were permitted to use, it had found similar ways of processing anomalous readings at three sites. It wanted to investigate a faint recurring temporal correlation: did it arise from the instruments, or from the data-processing program?

Lin Che asked what this had to do with complex spacetime.

“There is currently no basis for claiming a connection.”

“Then why is it on this list?”

“I proposed it.”

Lin Che read back. No one had assigned the task. While checking a dataset, the machine had encountered an unexplained residual, followed public records, and brought the three sites together. It had completed the checks that cost nothing extra. Continuing would require raw readings and some computing time.

Tess calculated the cost. It was not large. Clara wanted to participate, but Lin Che was waiting for her to finish an estimate for his own work. He said they could look at it next month. Then he noticed two equally uncertain tasks already scheduled because they belonged to his subject.

In the end they freed three days, with Clara to contact the data custodians. She said that if the work deserved continuing, more time would have to be budgeted. It could not count as a favour she did in the evenings. Lin Che agreed.

The first site opened its data quickly. The second offered only processed results. The third asked for the project lead's name. Tess entered her own name and contact details, but left Noah as the originator of the research question. The form came back once. She phoned to explain: that field asked who had posed the question; the person responsible for correspondence and costs was already listed below.

While awaiting replies, Noah continued checking the material already obtained. At the end of three days, two possible explanations remained indistinguishable. Lin Che read the results and authorised the task to remain open. He did not count it among the achievements of complex spacetime.

That year they finally found a result in complex spacetime worth publishing.

Previously, Lin Che had wanted to assign physical meaning to every imaginary component at once. Clara began with the narrowest class of models, separating changes that merely altered the representation from those affecting observable outcome probabilities. Autumn supplied a missing constraint. Noah expanded the candidates one by one and eliminated most of them.

The surviving models were still small and tightly restricted. Real and imaginary parts evolved together while preserving the requirements for real-valued observations and probabilities. In a restricted class of weak-gravity cases, they obtained deterministic motion. Under provisional observation rules, they also calculated the probabilities of signals at different locations in a simplified interference experiment, matching a known model. For the first time, Lin Che felt there was something between the ends of the whiteboard that they could continue calculating.

But the probability rule had still been inserted separately. They could not yet explain from complex spacetime itself why measurement had to yield those probabilities, let alone recover a complete quantum field theory. Lin Che wrote the limitation into the discussion. Reading it, he felt it made their best result look too small. The cursor rested at the end of the sentence.

Noah asked whether it should shorten it.

“Leave it.”

Independent verification took six weeks. Mira calculated by another route and found a discrepancy. Lin Che was uneasy at first. Eventually they traced it to different mathematical conventions. Only after expressing comparable quantities in the same convention did the curves coincide.

Clara listed the conventions. Autumn recalculated a page. On the video call, Mira nodded. Lin Che leaned back and released the pen he had been gripping. His fingers were a little numb.

That evening they set takeaway meals by the window. Hale asked what the paper would be called. Everyone offered a title and found everyone else's too grand. The final title was long and unremarkable. Tess said that if they printed it over the door, passers-by would assume they repaired instruments.

Lin Che opened his food and asked Clara to restore the conditions used in the derivation to the board. He wanted another look after eating.

He asked Noah whether Albert would have thought this path worth following.

Then he stopped himself.

“Never mind,” he said. “Check it first.”

Chapter Twenty Another Kind of Excellence

The 2036 paper brought Riverside Laboratory to many people's attention for the first time.

At the final draft, Lin Che added an extension after the finite-model results. This became the part that attracted most interest. They wanted complex spacetime to move beyond an approximation retaining only finitely much detail and answer a harder question: as more of the truncated field structure was restored, would the two methods of calculating probabilities still approach the same answer? This mattered if the finite models were to move closer to quantum field theory.

The probabilities from the two methods originally differed only slightly. Lin Che wanted to prove that, provided the state approached the prescribed conditions closely enough, the difference would vanish regardless of how fine the model became. The same experiment could produce different outcomes, but the two calculations should predict the same probability for each.

Noah ran many numerical tests. Clara organised the implementation. Lin Che completed the derivation. Hale read an early version and asked for clearer conditions. Autumn suggested publishing the finite case first and checking the extension further. Lin Che said he had supplied the estimate. She had to accompany a family member to hospital and missed the final discussion.

At a meeting before submission, Clara displayed the old error plot. As the model grew finer, the two curves still approached each other more slowly. Lin Che said he had proved that the delay would not alter the final result and directed her to the supplementary material.

She found the passage. “You say the error can be bounded. Does that guarantee survive when we restore more detail?” He moved a derivation to the centre of the table and said it should hold for the class of states they considered. Clara put a question mark in the minutes. During finalisation, Lin Che told her to mark the point as addressed. The question mark disappeared. The step was still not worked out.

In the second half of the meeting they had to decide whether to accept further funding. The funder required no particular conclusion, only a clear objective for the next stage. Tess said the contract could be smaller and expansion could wait.

Watching the two converging curves on the projection, Lin Che remembered waiting for payment in their leaking room. He did not want Clara to wait until the last moment every time her contract came up. He also wanted a place for several people they had already approached.

“Include the extension,” he said. “We've come this far.”

Afterwards Noah proposed further checks. He selected numerical tests they could finish before submission and marked the rest as future work. All the chosen tests passed. As Clara packaged the files, he stood behind her and felt that at last this could be put aside for a while.

After publication, Tess received several funding enquiries. People came especially to see the premises and ask about joining. Lin Che explained that the work had used university resources and depended on years of formal training. Reports still preferred the sharper story: a few young people outside the system had rediscovered what research meant.

He kept the reports. Sylvia later joined the board by invitation, helped find funding, and introduced the laboratory elsewhere. Once he asked her to remove “the results require further verification” from an introduction, saying that this applied to all research. She refused. He gave the journalist another version.

They moved into a building that did not leak. Tess had a proper desk. Clara began supervising two research assistants. Lin Che often wore formal clothes he had not grown used to. His mother visited in a coat she rarely wore. Before they entered, she straightened his collar, stepped back, and looked at him again.

“At least you won't need our dining table now,” she said.

She brought a plant and placed it by the window. Lin Che did not recognise it and asked Noah. The machine offered several possibilities and suggested a clearer photograph of the leaves. His mother said there was no need for so many questions: water it when it dried out. A little later she added that he must not water it too much either.

He introduced everyone and explained what they could do here and what would go over there. She understood little, but greeted each person. At the group photograph on the wall she recognised Mira. Lin Che was surprised she remembered. His mother said that for a whole year he had called every day about the teacher with the broken nose pad. How could she forget?

He laughed and meant to say more, but his phone rang about the next lecture. His mother sat by the window with a beautifully printed booklet, turned a few pages, and started again at the beginning. Through the glass he signalled that he would be a moment. She nodded.

After the call there were emails. When Lin Che emerged, she stood and said she should go. He had promised to take her home. At her words he looked first at the time. She had already folded the plant's paper bag, leaving only a container of food.

“I know where the stop is.”

After the lift doors closed, he opened the container. The food was cold. He messaged her to let him know when she got home, then reopened his slides. Much later his phone lit up with “Home now.” He saw it and meant to reply after finishing the slide. When he finished, he forgot.

The slides had originally separated complex-spacetime assumptions from proven results. Later he combined the two pages to leave time for an introduction to the laboratory. At the end someone always asked whether they had proved randomness merely apparent, or unified relativity and quantum field theory. He always said no conclusion could yet be drawn. Back at his computer, he was nevertheless pleased to see the rising traffic.

They became known for allowing failure, and gradually learned how to display failure to visitors. The favourite example was the pair of curves in Lin Che's seventh file at sixteen. Hale's two complex numbers, written on the back of a meeting notice, filled a large screen. At the point where he described believing he had touched the universe's secret, Lin Che paused. The audience always laughed there, and he turned the page, reassured.

Noah asked whether they should include the still-unresolved error plot. Lin Che said this audience was not specialist; the explanation required too much background. The next audience was specialist, but then he felt the plot belonged to work in progress and should not be explored in a lecture.

Meanwhile, Noah's calibration question received a response. At first the three datasets appeared to corroborate one another: each site's research system had found the faint correlation. When the second site eventually supplied raw data, Clara checked it alongside the first and discovered that all three processing descriptions referred to the same program. The two supposed repeat checks had also called the same version of that program.

The program had shortened an interval that should have been retained. Bypassing that step removed the most conspicuous correlation. The third site's raw data remained unavailable, so they could not conclude that the same was true there. Clara removed a chart labelled “Three Independent Verifications,” replaced it with the three actual processing histories, and wrote to each custodian.

Some thanked her; some asked only whether published papers were affected. One shared platform would not change its automatic verification badge without a runnable example. Noah produced one. The platform then required confirmation from a person who had actually checked it. Clara spent Saturday by the window rerunning the example. Her lunch went cold before she finished it.

Another residual discrepancy remained unexplained and required new data. The instrument's time was already reserved by long-term partners. Buying three nights at short notice exceeded the money set aside. Lin Che offered to write an application. The facility agreed to place it on a waiting list. No one promised a date for the experiment.

Felix Ward joined them too. He was the student who had first asked how many years a result would take at the advice session. Now he calculated complex-spacetime models. After eight months the results were still unclear. Lin Che asked what he had learned. Felix said he did not know. The meeting room fell silent. Lin Che asked again: surely his understanding had changed in some way?

Felix glanced at the renewal form in front of Tess and said he would put something together that afternoon.

The report read smoothly. Every failure led to the next step; the previously disordered attempts acquired coherent reasons. Lin Che finished it with relief.

At filing, Noah noted that it had helped prepare the report. The original contained only a few lines: I don't know what to do next. I can't see what these calculations mean.

Lin Che compared the original and revised versions several times. The renewal form remained blank. The laboratory had fourteen people now. “We allow failure” could not be a promise that everyone could stay forever.

Later Felix taught a new assistant how to fill in the form. He explained patiently which attempts could be grouped and which detours should be recast as the elimination of alternatives. He had examples for everything.

Passing by, Lin Che heard: don't simply write that there was no result; explain the value produced by having no result.

It was a sentence he had used at a meeting himself.

They designed a new assessment form, removing paper and citation counts and adding originality of the question, capacity for sustained inquiry, and the value of failure. Noah

standardised applications into summaries so that reviewers could compare them within limited time.

Noah marked passages that did not fit the categories and recommended reading the originals.

Lin Che read them. He added a line to the procedure: consult the original where necessary.

At the meeting, only summaries appeared on the screen. Someone asked whether the originals had been checked. Someone else said the person compiling them presumably had read them.

The reviewers were still mostly people skilled at obtaining funding and familiar with the language of reports. Tess explained that at least they knew how to choose when money was short. Lin Che did not object. His own name headed the list.

Someone who had seen Felix's original few-line report judged him to have “insufficient capacity for sustained inquiry.” The same work, presented through the polished report, scored well at formal assessment. Tess collected the renewal papers, and the meeting moved on.

They were invited to explain their new assessment method. Tess sent the forms and program documentation to several partner groups, asking each to designate readers for the originals. The first to reply had only half a clerical post available and asked whether it could temporarily rely entirely on summaries. They discussed it once without sending a formal answer. A month later a screenshot arrived: the form was in use, with Riverside Laboratory's name still in the footer.

Lin Che complained to Sylvia that it resembled Gresham's law: bad work driving out good. She picked up two papers on the desk and asked which was bad.

He could not say. Both were correct. One had saved other people considerable trouble.

“But if both count as one paper, and one takes three months, who's going to stake five years?”

Sylvia put them down and asked how much uninterrupted time his own laboratory had devoted to the old error plot that year.

He wanted to say they had been looking at it all along. Opening the diary, he found that several dedicated discussions had become preparation for talks.

Wednesday afternoons had originally been reserved for open discussion. In the first month Lin Che arrived late because of meetings. In the second he asked Noah to chair the sessions until he could read the record. The machine organised the questions thoroughly, even ranking them by relevance to the main complex-spacetime programme.

Once, when he finally sat down on time, everyone first asked whether they should follow the summary's order. The whiteboard was clean. Fourteen chairs stood round the table. He invited them to speak freely. No one spoke. In the end they began with the first item in the summary.

Later Clara was offered a fee to teach applicants how to present valuable failures. She declined.

Lin Che asked why.

“I'm afraid all they'll learn is how to make failure sound good.”

He smiled, saw that she was not smiling, and slowly stopped.

Chapter Twenty-One Page Four

Two years later, Lin Che saw Owen Easton's name for the first time.

He was forty-six and taught mathematics at a secondary school in the northern suburbs. His research record had a nineteen-year gap. The attachment ran to eleven badly scanned pages, one marked by half the base of a cup. He was asking for two research days a week. The system's summary said he wanted a full-time position.

Lin Che did not notice the difference.

Autumn recommended including him in the final discussion. In the original she had seen a brief objection worth asking him to explain. Lin Che read the first and last pages, then the summary. He thought it too close to existing research, with no clear independent contribution.

Only two places remained. He had a lease meeting that afternoon and a fundraiser that evening. Two other candidates had already been waiting three months.

Noah generated a rejection thanking the applicant for maintaining his enthusiasm for science over many years. Lin Che removed a compliment that sounded patronising and prepared to send it.

Then a new email arrived.

“I don't necessarily need a position. There may be a problem in your paper from two years ago, at the step extending the model with finitely much detail to the full case. I've drawn a table on page four of this attachment.”

Lin Che opened the scan.

It was a grid. Each row represented a level of model refinement: farther down, more detail was retained; farther right, the state approached the prescribed conditions more closely. Owen had drawn arrows to the right along each row and a red diagonal slanting downwards to the right.

Lin Che looked for a long time, stood, and moved the coat stand behind the door to uncover the whiteboard.

They had proved that, at a fixed level of model refinement, bringing the state closer to the prescribed conditions brought two probability estimates for the same measurement outcome closer together. The paper went on to say that this remained true even as more detail was restored. A guarantee valid only in the finite case had been carried into the later conclusion.

Noah first showed a simple example. As long as no further detail was added, a smaller deviation produced a smaller discrepancy. But along Owen's diagonal, the model grew finer while the deviation shrank, and the final discrepancy remained.

This was not yet a counterexample to the complex-spacetime model. It merely exposed an invalid general inference. Lin Che's first thought was to wish the example were more complicated.

At least then he could explain why he had missed it.

He asked Noah to return to the model itself. If the troublesome states never satisfied the physical conditions, perhaps the conclusion could be saved. The machine filtered some out. It could not exclude all the rest. They met the constraints stated in the paper; their growing effects accumulated in the details previously truncated away.

When Clara returned, she checked with a different program. At the first results, she glanced at the time and switched off her phone. Hale came that night with one button of his jumper fastened wrongly beneath his coat. The three took the paper apart, verifying it passage by passage.

The same initial conditions, the same measurement arrangement: as details were restored, the two algorithms still produced different probability distributions. The difference they had expected to vanish did not vanish. Lin Che had claimed that the model produced probability predictions independent of the computational method. Now the counterexample lay before him.

The finite-case results still held. The much-publicised extension could not retain its original claim.

At one in the morning Clara said she needed to go out for a walk. Lin Che thought she was leaving and stood to explain. She shook her head and took only her access card. Some fifteen minutes later she returned with three rice balls, saying the convenience store had nothing else.

He opened his rice ball. Several times he wanted to ask Clara what they should do next. She was examining new results. Afraid to interrupt, he ate half and left the rest beside the desk.

Lin Che asked for all the old records.

Noah found a note made sixteen days before submission: continued control of the error as the model is refined remains unproved; existing numerical tests cannot replace a proof. Beneath it was Lin Che's reply: organise under the current assumptions for now; put the relevant limitations in the discussion.

Before submission he had moved the limitation into the discussion, leaving only “more general cases warrant further investigation.” The words “remains unproved” had disappeared.

“Do you remember?” Clara asked.

He said not clearly. She enlarged the note. He leaned closer and recognised his own reply beneath it.

At the time he had thought one final check would fill the gap. Hale believed the younger researchers had done so. Clara had taken “addressed” at his word. The derivation had remained until this letter arrived.

Clara opened the minutes of the pre-submission meeting. “That was my question. Does the guarantee still hold after more detail is restored?”

Lin Che remembered. The afternoon had been crowded. His hand had already been on the next-slide key while he answered.

“I said it had been addressed.”

Clara put away her uneaten rice ball and asked which part they should check first. Lin Che told her to go home and sleep while he found every original file relating to that step.

In the early hours he wrote to Owen, asking him to come the following day.

Owen said he taught during the day but could come at half past six. He also mentioned that he had already sent three letters and wondered whether he had used the wrong address.

The next day Tess found the records. The three emails discussing the paper had been classified as general enquiries and placed in a queue awaiting technical review. The person sorting them had followed the order Tess approved: material from familiar institutions went for review first.

Owen's first letter identifying the problem had arrived three months earlier.

Lin Che closed his email and suddenly snapped at Noah. “You have the letter Planck wrote to Einstein. A schoolteacher writes to us, and he has to wait three months. Why?”

The three receipt dates appeared on the screen. Every letter was there. Two had been merged into the same enquiry. With the earlier one unread, the later ones received no new place in the queue.

Noah displayed “consult the original where necessary” from the assessment procedure. Lin Che recognised the words he had added. Once they had seemed a small allowance for overworked people. Now anyone could use them to pass the original manuscript to the next person.

Tess pulled up a chair and asked who decided what was necessary. Lin Che meant to say the sorter, then opened the permissions. That person could change labels but not priorities.

“First export the processing history of all three,” he said.

They followed two automatic classifications and one human confirmation, ending at the waiting rules the laboratory itself had approved. Nowhere did it say a schoolteacher's letter must not be read. Material from familiar institutions simply kept going ahead of it. Tess opened the version they had distributed and asked who else had received it.

Lin Che found seven addresses. Two recipients had forwarded the instructions to other partners. He could not immediately say who was using them.

He told Noah to disable the automatic replies and reopened Owen's attachment. This time the seven addresses remained beside page one.

Chapter Twenty-Two After School

Owen arrived at six forty. At the door he apologised: there had been a problem with the school bus, and the last pupil's parents had not collected them. He could not leave first.

He carried a black backpack with a paperclip replacing the broken zip pull. Lin Che recognised the design; he had used one at secondary school, though this one had been washed almost grey. After sitting down, Owen first took out a stack of examination papers, then found the folder containing his manuscript.

“I've redrawn page four. The old one wasn't very clear.”

“It was clear enough,” Lin Che said.

Owen looked at him, took this for politeness, and pushed the new drawing across anyway.

Lin Che told him they had confirmed that the step did not hold and were checking the extent of the damage. He apologised for not answering the earlier letters.

Owen nodded and looked for a pen. Lin Che had prepared an explanation of the queue and summaries. Seeing the other man already circle a symbol, he said no more.

They talked for more than two hours.

At first Lin Che still found himself explaining the overall complex-spacetime proposal. Owen listened, drawing a short arc in the margin. When Lin Che finished, he asked precisely which states were permitted after the omitted detail was restored.

Lin Che turned to the definition. Owen asked whether, under that clause, the counterexamples they had just discussed were still included.

They were.

“Then let's start with those.”

Pointing to that class of states, Owen asked Lin Che to work forwards: how the real and imaginary parts coupled, at which step the probability of an observation arose, which difference should eventually vanish. At every step he asked him to pause and add the necessary conditions in the margin.

They soon filled the blank sheets. Owen took spare exam papers from his bag and turned them over. As one page flipped, Lin Che saw a question on the front: Discuss the conditions under which the function is continuous.

The second half of Owen's manuscript attempted a repair. Part of its proof was wrong too. When Clara pointed it out, he frowned, worked a few lines, and said, “Yes. My estimate doesn't keep the error under control either.” He crossed out the passage and asked whether another estimate could be used.

Lin Che had begun stiffly. As they continued, he reached across for Owen's paper, saying “Wait” twice. Owen waited. Lin Che bent to supply a missing step, then pushed the page back.

They stopped only when Owen's phone vibrated.

He went outside to answer. Through the glass, Lin Che saw him raise two fingers as though the person on the phone could see. On returning he capped his pen and said he should go: he had promised his child two pots of yoghurt.

“How do you find time to read all this?” Clara asked.

“I don't, much.”

The school had introduced a course in AI applications the previous year. Preparing to teach it, he had reopened some old books. Later he read their paper and could not follow one step. He asked the teaching platform to explain. The answer was fluent, but still seemed wrong. In his first letter, he suspected that after so many years away from research he had overlooked something everyone else knew.

“And then?”

“Then I kept looking at that one guarantee. Why wouldn't restoring more detail cause a problem?”

At first he read during the pupils' lunch break, taking several sessions over a page. Physical meanings he did not understand were marked for later reading at home. The teaching platform's explanations were sometimes useful; sometimes they merely recast the original in smoother sentences. Believing the mistake his own, he copied the same page out three times.

He had only wanted to understand that step. But on the next page the same guarantee suddenly applied much more widely. He was always telling pupils to check the conditions in a problem, yet he too had nearly skipped them. So many people had already cited the article.

After sending the third letter, he stopped for a while. He had end-of-term papers to prepare and things to do at home. He wrote a fourth time because his daughter put a glass of water on the manuscript. Pulling out the pages, he saw the grid again, thought for a moment, and rescanned it.

Lin Che looked at the half-circle on the scan. The first time he had seen it, he had frowned.

He asked whether Owen still wanted to come and do research.

“Yes. But first I need to know what you want me to do. I can't give up all my classes.”

“You'll receive proper credit and pay.”

Owen waved a hand. “Credit the correction according to the usual rules. A job is another matter. Whether I can do it depends on what comes next, not on this letter.”

Lin Che asked whether he would have stayed in research if someone had given him five years earlier in his life.

“Back then I couldn't even settle where I was going to live,” Owen said. “For now, could we begin by freeing up my Thursdays?”

Lin Che brought a calendar and asked which Thursday they should start with, and whom they should contact at the school.

At the door Owen stopped again.

“If someone wants an interview, I don't want a story about a schoolteacher defeating artificial intelligence.”

Clara said she did not want that either.

After seeing him out, Lin Che returned to clear the table. The exam papers were still on a chair. He hurried outside and just caught sight of a figure with a backpack across the pavement, bending into a convenience-store fridge.

He photographed the papers and messaged Owen that he could bring them the next day.

The next afternoon he waited at the school gate. Pupils passed, talking about entirely different things. Owen came out, took the papers, and leafed through them. He found the one with their derivation on the back and said they would still need it.

Lin Che offered to buy him dinner. Not today, Owen said: parents' evening. They exchanged a few more words about the problem by the fence. The bell rang. Owen nodded and returned with the papers.

On the journey back, Lin Che opened his complex-spacetime notes. On the earliest page was a clock and his old question: What, exactly, does it read? He turned a dozen pages, then returned to it. Beside the old date he added a new one without crossing anything out.

At the laboratory Noah was still on. Their jointly annotated manuscript filled the screen, black text crowded by red lines and comments in different colours. The machine had organised the confirmed parts and listed three points for further checking.

“Show me Planck's letter,” Lin Che said.

The old image appeared. Some handwriting was unclear, and Noah offered two possible transcriptions beside it. Lin Che paused at the line asking about the derivation's conditions, then looked again at the pages Owen had left.

He closed the image and added Owen's name and the dates of his letters to the correction draft for his co-authors. He moved the admission that the original argument failed to the beginning.

He also wrote that he would take responsibility for answering researchers who had cited the work. Clara read it and listed several groups they needed to contact first.

Chapter Twenty-Three Those Who Left

The board met on Friday.

Tess gave everyone a cash-flow statement. The first page paid the expenses from confirmed funds only. The second included grants still under application. Lin Che looked at the shortfall at the bottom of the first page. It was larger than he had thought.

Someone suggested completing the technical investigation before explaining publicly. Clara said a few points still needed checking, probably several days' work. The person asked whether they could wait until the next funding round was confirmed.

“We can wait for details,” Autumn said. “We can't let the next application keep citing that conclusion.”

A board member asked whether the original finite-case results remained valid.

“Yes,” Lin Che said.

“Then we can't say the whole project has failed.”

“I haven't said that. What we need to withdraw is the part we claimed to have solved and hadn't.”

The man sighed and laid his glasses on the paper. He had introduced two funders to the laboratory and would now have to explain to both. Lin Che had been growing irritated. Seeing him write the two organisations' names on the sheet, he did not interrupt.

The board member asked whether stricter conditions on complex spacetime would let them continue.

“We can investigate which conditions help,” Lin Che said. “But we can't remove every state that breaks the conclusion and then pretend we haven't added anything.”

The man circled a sentence in the correction he did not understand. Lin Che explained it twice, then shortened it: as the model becomes finer, the two algorithms predict different probabilities for the same experiment. The board member read it again and asked how many applications already sent still described the extension as a reliable result.

Another member asked whether, since the paper had passed peer review, they should first await a decision from an expert panel.

Hale turned Owen's grid towards him. They could seek independent checks on the extent of the problem, he said, but a vote could not replace the counterexample already on the page.

“We can decide whether to continue funding,” Sylvia said. “We can't decide whether the probabilities from the two algorithms coincide.”

She had joined the earliest funding discussions and recommended the laboratory in public. After speaking, she added her own name to the list of people who needed to explain.

Lin Che looked at Owen's grid. He had said in lectures that nature had the final vote. Now the conclusion he had to withdraw had not even survived its own derivation. He turned the correction back to the first page.

Halfway through the meeting, he offered to stop his own salary first.

Tess did not immediately agree. She showed him the first cash-flow page: his salary would fill only a small part of the gap. How would he pay rent?

“I can take more outside work.”

“Fine. Write down how much you can take, and how many hours a week that leaves.”

She turned the sheet and continued through fourteen contracts. Two people had just renewed visas. One person's partner was undergoing treatment. Someone else had signed a lease on the strength of a renewal indication. The reserve could pay the agreed transition costs. Rent and medical bills would continue afterwards.

Lin Che listed the work he could accept, then checked the contracts with her one by one.

The staff heard about the correction and funding risk before the media did. Some asked questions; others said nothing. A young colleague asked why, when they had joined the previous year, the laboratory had confidently promised at least three years' support.

Tess explained the budget as it had stood. Lin Che said the funding had not all been secured; he had promised too much while recruiting.

The colleague listened and looked down at a phone. After that day they never again discussed research alone together.

The correction went to the journal and onto the laboratory website. The relevant original records and revision history were made public alongside it. Owen's name appeared in the correction, followed by the date of his first letter.

For the first few days Lin Che's phone rang constantly. Some praised their honesty. Others said small institutions were predictably unreliable. Soon the reports dwindled and journalists pursued other news. The journal editor still asked for details. Funders' letters continued arriving. The office itself had grown quieter.

Harder to answer was an email from another research group. They had spent eight months following the extension, and several pieces of work now needed repeating. Lin Che explained the affected areas point by point and offered verification records that were not yet fully organised. They neither scolded nor consoled him. They listed six questions requiring answers.

He put those six questions on the left of his screen and stopped preparing a new article. Noah asked whether another task should be postponed. He said yes.

That week the seven addresses also began to reply. Three said they had stopped using the procedure. Two still awaited management approval. One asked whether Riverside intended to withdraw the whole assessment method. The last asked them not to identify the organisation publicly yet: the applicants concerned did not know how their material had been processed.

Lin Che spent an afternoon revising replies. He had to explain separately what the summaries had misread, how the ranking rules prolonged waiting, and what remained unknown. “We have made improvements” would not tell recipients which files needed another look.

Tess received processing logs from organisations that agreed to review them. One materials researcher could work only at night. The system had summarised “I cannot guarantee daytime availability” as insufficient commitment of time. Her calculations contained errors on review; the

summary's judgement about her also needed correcting. Lin Che wrote the two points separately and contacted her.

She declined the invitation to reapply. She had changed jobs and cared for family in the evenings. She asked only that the correction remain in her original record, so that the judgement would not be reused next time.

Another group recovered an overlooked attachment and found that the applicant had identified an already-solved problem. They still wrote to explain why there had been no reply. One organisation never supplied its processing logs. In the public account, Riverside could only mark the matter as unknown. It could not claim to have checked on that organisation's behalf.

The hours quickly exceeded the budget. Someone suggested classifying the work as assistance to applicants. Tess refused. She recorded it as the cost of correcting the old procedure and moved money from activities that could wait. The entry was not for a single talk. Someone would have to keep answering calls over the following weeks.

His mother heard through an article someone forwarded. When she called, Lin Che expected to explain again and began by saying not everything had been wrong. She waited until he finished, then asked whether he was coming home that week. He had not planned to. In the end he said yes.

It was late when he arrived. He inserted his key, but the lock would not turn at first. He drew it slightly out and the door opened. The lock had been like this for years. He had changed access cards several times; the key in his hand remained the same.

His mother looked out from the kitchen and told him to wash his hands. Two places were set and the food was covered. Once he sat down she told him the pool had a new instructor and asked whether he had been eating in the office. Several times he failed to hear. His eyes kept straying to her phone, face down on the table.

At last he put down his chopsticks and began explaining the difference between the finite model and the extension. She tried to follow, then asked whether some of what they had got right before could still be used. He said yes and immediately added qualifications. As her brow tightened, he spoke faster.

“Xiao Che,” she interrupted.

He stopped. Apart from her, almost no one called him that any more.

“I still don't understand. You don't have to explain everything tonight.”

He looked down at the rim of his bowl. After a long time he said he might have cost several people their jobs. Some errors had been pointed out earlier; he had not read the warnings properly. When recruiting, he had promised people several secure years to work. At the time he had truly believed he could provide them.

His mother laid her chopsticks flat on the table, listened, and asked whether those people knew now.

“I've told them. But there's still a lot to do.”

“Then you have to keep doing it.”

He nodded. She asked whether the money for the next payments was in place. Slowly he explained the reserve, salaries, and contracts. These things she could understand.

When he finished he leaned back, too tired even to lift his cup. He told her he would need more software work and the laboratory might move somewhere smaller. He had meant to reassure her. However he calculated it, the second half of the year remained uncertain.

“You can live here,” his mother said. “I can't help much with the rest.”

“At my age.”

“You still need somewhere to live.”

Lin Che picked up the cup without drinking. Sometimes, he said, he was afraid even to call her. The last few times she had asked whether he had eaten, he had been discussing cuts with people. Only after hanging up had he realised he had eaten nothing all day. He was afraid she would hear it as soon as he spoke.

She was silent for a long time. A car passed outside, its light moving across the wall.

“You don't have to find good news before you call me.”

He could not speak. The cup remained in his hand. She came over and rubbed his shoulder twice. He took her hand. Along its edge was a blue stain she had not quite washed off, left by writing during the day.

“Mum, I really did get this wrong.”

“I heard you.”

He lowered his head. As tears fell onto his clothes, he still wanted to finish his explanation: not all the work had been wasted; he still had ways forward. His mother did not wait for those words. She sat beside him and let him rest his forehead against her hand.

She cried too. She still did not understand where the paper was wrong or what remained useful. She knew only that he was gripping her hand too tightly. She remembered him as a child, too sleepy to sit up, a hand still pressed on his book, begging her to leave the lamp on a little longer. With her free hand she reached for the tissues.

Lin Che took one, wiped his face, and asked whether she was disappointed.

“I don't understand those things you do,” she said. “When it worked, you were so happy that I was happy too. But seeing you like this, am I supposed to wait until you've done something else before I let you come home?”

He shook his head. She carried the cold dishes to be reheated, then turned at the kitchen door and asked whether he was staying that night.

“Yes.”

He slept in his old room. Several of his mother's storage boxes stood on the wardrobe. The desk was still there; the lamp had a new bulb. Unanswered emails remained on his phone. He glanced at them, turned the screen face down, and fell asleep.

In the morning his mother went swimming as usual. By the time he woke she had left breakfast in the kitchen. He sat at the dining table, opened the six questions, and began with the

first. At one point he wrote a long explanation, then deleted it. He stated plainly that he had not checked that part before and attached the records that could be examined.

When she returned in the afternoon, he rinsed her goggles and left them by the window. Before he left, he wanted several books. From a cupboard she took the first paper he had brought home as a university student and asked whether he still wanted it. He opened it. The conditions were written out one by one, so cautiously they seemed almost clumsy.

He put it with the books. At the door she asked whether he would come next week. He said he would, and would call first if anything changed.

They did not renew the lease on the new building.

By the following spring, seven of the fourteen had left. Some found better positions. Others simply had to find a position. The reserve was paid out according to the contracts. Some still resented them. Lin Che no longer asked these people to believe in the laboratory's original intentions.

Felix went into data services. Before leaving, he kept the report on the value of failure in the shared directory and sent another set of unpolished notes. They contained several points he had never understood. One concerned the very condition they later checked repeatedly.

Lin Che began asking why he had not said so at the time, then deleted the message halfway through. He found the meeting recording and heard himself urging Felix to begin by summarising what could go into the progress report.

He replied point by point with what he could currently answer. The following day Felix said only that he had received it. He made no promise to continue the work.

Clara left too.

She joined an engineering company to verify models. Some work was dull, but the salary arrived monthly and leave was clearly defined. Before going, she handed over the programs, required files, and unfinished checks three times.

Lin Che asked whether she would still come back for discussions.

“Yes. But don't put my evenings in your budget.”

He nodded.

She laid her access card on the desk, reconsidered, and took it back. “I still need it today. Let me get the boxes downstairs first.”

The final handover page listed three questions she still considered worth pursuing. At the second, Lin Che asked why it had not been marked as a priority.

“Because you wanted to finish that paper first.”

He remembered what he had said and drew the page closer. Clara did not explain further. She packed the contract she had signed when she first arrived. The two handwritten amendments were still there.

They carried two loads together that day. The lift stopped working for a while. On the stairs, the bottom of a box wore through, spilling charging cables and a lunch container. A mug

chipped. Clara picked it up, felt the edge, said she had used it for years, and put it back in the box.

Autumn still came on Saturdays when she could. Her mother needed care, and she had taken another consulting job. After retirement, Hale brought a desk the university was disposing of. An adhesive mark on the top would not come off. They offered to buy him a new one. He said this was enough; the drawers were excellent.

The school could not immediately rearrange its timetable. Owen's first contract was for six months and two evenings a week. Pay covered reading and checking and did not depend on new conclusions. He first completed the counterexample to the lemma, then discussed with Lin Che whether a weaker conclusion might survive.

They added a restriction to the previously omitted detail, ensuring that its cumulative effects remained controlled. With this extra condition, the two methods could continue approaching the same probability predictions as the model grew finer. Why that restriction should hold, and whether the probability rules could be obtained from complex spacetime itself, remained unanswered.

Lin Che tried to derive the restriction from the model's own dynamics. Overnight Noah produced more than a dozen candidates. The next day they eliminated nine; the rest each had difficulties. The most attractive merely changed the symbols of the condition needing explanation and returned it to the starting point.

He compared the two versions term by term and finally laughed: after all that, it was the same condition. He told Noah to retain the derivation, state the limitation clearly in the abstract, and then worked with Owen to complete the weaker proposition.

Autumn read it, corrected one notation, and asked who would bring tea next week.

A foundation that had considered supporting expansion decided against funding. Sylvia brought several co-funders together and reopened the five-year plan that had gone unrealised for years. This time they began with the sums each person had already committed, then decided how much time those sums could buy.

One contributor asked why they should offer five years precisely after the researchers had got something wrong.

Sylvia produced two reports. She had asked researchers uninvolved in the original paper to read it afresh, assess what survived, identify difficulties still worth pursuing, and judge whether Lin Che and his colleagues could continue.

The reports disagreed. One found the overall complex-spacetime direction too optimistic. The other was interested in its geometrical structure. Both were willing to read the weaker proposition. Both objected to publicity claiming that determinism and randomness had already been unified.

Sylvia brought each passage to the meeting. Someone asked what result would be required after five years. She said she could not guarantee a result either. “These two people have read the work. Their reasons are here. For these people, on these questions, are you willing to provide five years?”

At first Lin Che, sitting beside them, kept wanting to speak. Then he noticed that both reports discussed a passage he had become accustomed to writing. One reader identified a difficulty more serious than Lin Che had ever admitted. He put away his prepared introduction and made a note.

The money covered only two part-time awards, modest computing resources, and fees for readers. It could not restore the fourteen-person laboratory. Five years of funding was reserved separately, without annual payment decisions based on paper counts. Researchers would submit yearly work records. They could change questions, but had to explain why.

Lin Che asked what would happen if the path ultimately failed.

“Explaining why it fails can be useful too,” Sylvia said. “But don't rush to decide what five years will bring.”

The agreement also allowed researchers to withdraw. If they changed their place of collaboration, the funders would consult other peers, and funding could move with the researcher rather than remain at Riverside. Owen's first question was whether he could keep teaching at the school.

The first award gave Lin Che two days a week. He still needed his software work. The second supported Owen for fewer hours and enabled the school to adjust his timetable. Tess checked twice before marking the Thursdays.

Sylvia brought the closing report from another grant. After years of work, the project had not obtained the hoped-for result. It had established only why several common approaches failed. The report was short. She brought it unchanged.

“This one wasn't a success,” she said. “My judgement was wrong too. But someone has taken up another project this year and used it somewhere we never imagined when we funded it. One or two out of ten like that would make it worth trying. The rest still need honest accounts. We can't call them all promising for the future.”

She attached the report to the budget.

Tess kept the restricted grant on a separate line. Rent and everyone else's pay still depended on the paid projects waiting behind it.

They moved back to the original room. The new long conference table would not fit. They sold it to the company downstairs for less than half its cost. The electronic display by the entrance was removed too, leaving four dark holes in the wall.

Before surrendering the lease, Lin Che made a final check of the emptied new rooms. On the windowsill he found a page from an old brochure. In large type it said: Let No Important Idea Go Unheard.

He did not tear it up. He filed it with the lease documents. On the back was a budget he had written, its headcount revised twice.

Tess waited outside for him to return the keys. In her hand was the router they would need the next day.

“Come on,” she said. “The internet isn't connected over there yet.”

Chapter Twenty-Four Unfinished

In the year Lin Che turned thirty, the laboratory still received more letters than it could read.

Sylvia named the grant the Long-Term Research Partnership. At the first remote discussion, the system classified Owen as non-research staff. He left “secondary-school teacher” unchanged and entered the calculation he wanted to discuss. Tess moved everyone's specific work ahead of their job title.

Lin Che uploaded the reality conditions and several probability constraints to an open problem network. Others continued checking them and left revision histories. A research cooperative took turns verifying its members' calculations. Clara's company set aside time for public discussion, and she joined from her work computer. Another voice came from a bedroom; halfway through, its owner got up to switch off a washing machine.

One new verification had no corresponding presenter. It came from a research system at a public computing centre. The system had independently selected the task from an open problem repository and, after several rounds of calculation, changed the range of states it examined. The duty staff confirmed only that the records were complete, not that the conclusions were correct. At first Lin Che looked for an author, as he did when checking letters. Eventually he spent the time on the supplied execution logs. At a passage he did not yet understand, he asked Noah to check by a different method.

Hale remained in contact with the university after retirement and helped find literature and computing resources. Lin Che took a measurement proposal to experimental researchers. Understanding the noise sources alone consumed an afternoon. Back at the laboratory he removed a promise about precision and asked Hale to arrange another discussion. Maintaining instruments and preserving data for years were things he could not afford in the little room. Several instruments were booked into the following year. He noted the dates and requested a preparatory meeting first.

Half a day each week went into the work schedule for reading correspondence. Whoever was on duty handled that batch. Noah first helped classify the material, check calculations, and find related work. Letters identifying specific problems in existing research went into a separate queue and notified the relevant authors directly. The queue was publicly visible so correspondents could see their place.

The system still had many difficulties. People submitted the same manuscript repeatedly under different names. Sometimes the duty reader was absent. Sometimes they argued over a letter for hours only to discover a miscopied symbol.

Revising the budget, Tess asked whether half a day could become two hours. Her reason was practical: that month's paid projects were already overrunning.

Lin Che almost agreed. He was revising a client's program when a progress enquiry appeared in the corner of his screen. In the end they negotiated an extra day for delivery and hired an external reader by the hour to share the letters. They would still have to find next month's money next month.

When a second group of candidates submitted applications, someone proposed scoring “a sense for important questions.” Someone else produced a writing guide based on successful previous applications. Sylvia sent the material to Lin Che and asked him to review it.

He looked at it for a while, then asked whether there was time to read the originals. Each application had fifteen minutes, he was told, and the system would prepare summaries. With another client deadline that week, he nearly filled in the assigned scores directly.

Instead he accepted only two applications and asked them to find readers for the rest. This delayed the assessment by a week. A candidate asked about the date. Lin Che explained in his reply and left an address where he could be contacted directly.

One of his two applications was exceptionally fluent. At the end, he still did not know where the author was stuck. The other attached only a failed example. He wrote questions for both and invited them to explain. In the second person's calculation, he also circled a possible error.

Noah was awaiting a small renewal fee too. For its earlier calibration question, they had eventually borrowed an instrument at night. Clara and a technician elsewhere repeated a set of measurements, ruling out one explanation. One remaining discrepancy persisted. The machine proposed changing the readout method, which required more rented computing resources.

Looking at the budget, Lin Che habitually asked whether it could first help them finish the next complex-spacetime step. Noah listed the missing conditions for both tasks. It supplied no guaranteed completion date.

“One has to come first,” he said.

“They can be ordered according to your current objective.”

Lin Che remembered that he had not chosen this question in the first place. Tess asked how they would respond if another collaborator had proposed it. He gave both questions to Autumn. In the end they reserved a little time for each and stopped a calculation added only to improve a report. Noah had not received unlimited permission. Lin Che's work no longer automatically came before its own.

One rainy afternoon in May, it was Lin Che's turn to read the letters.

He began with two invalid proofs and identified the precise faulty line in one reply. The third letter came from someone who maintained warehouse equipment and had attached hand-drawn circuit diagrams. The correspondent used complex numbers at work, which had sparked an interest in complex spacetime. If a complex number could be split into two real numbers, the letter asked, what had actually been added by changing the representation?

In the attachment, the writer assigned every imaginary part an as-yet-unseen direction, then discarded it when calculating a reading. Since it could not yet be measured, the writer thought there was no harm in leaving it out of the result for now.

Noah opened Lin Che's seventh file from when he was sixteen.

“I remember,” he said.

The coincident curves were still there, “successful” crossed out beside them. More than a decade later, complex spacetime was still not a complete, testable theory. They had obtained restricted results and found counterexamples. Whether the deterministic evolution of general

relativity and the probabilistic outcomes of quantum field theory could arise from one set of complex-spacetime laws remained the original question before them.

Sometimes he thought this was progress. Sometimes, faced with the lengthening conditions, he wanted to remove them all and start again.

At the end, the correspondent asked whether complex spacetime predicted a new phenomenon and whether the available instruments could detect it. Lin Che copied the question onto a fresh page. Noah listed several directions. Against each, he marked what was established and what remained an assumption.

He began his reply: Thank you for your interest. This idea has already been extensively studied…

He stopped and deleted the sentence.

He typed out the example Hale had once written on the back of a meeting notice, explaining that the prematurely discarded part also affected the result. But adding an imaginary component to spacetime would not automatically produce quantum probabilities. Why should it evolve this way? How were the probabilities of different outcomes obtained? Under what conditions would it recover relativity's deterministic description? Each step needed a reason. So far they had answered only some of them.

He could not currently propose a new phenomenon that was both sufficiently well founded and suitable for testing with the correspondent's equipment. He said so plainly and attached two references. One was their correction.

The email was long. He removed two technical terms and added at the end: If you find a mistake in the example, please write again.

After sending it, he saw that only twenty minutes of his half-day remained. Nine letters were unread.

“Continue?” Noah asked.

“The next letter.”

The rain thickened. Hale's old desk stood by the window, its adhesive mark still visible beneath a thin sheet of glass. Owen would come that evening. Clara was due to review new verification results the following week. Autumn had taken time off. Tess was out discussing a leak with the landlord. Lin Che was alone.

His mother's plant had grown large. A branch had broken during the move and Tess had propped it up again. An expired event poster lined the space beneath the pot. Lin Che touched the soil, remembered he had watered it, and put down his cup.

He saw a photograph his mother had sent. She stood beside the pool with several classmates, each wearing a very unflattering swimming cap. He enlarged it and recognised the person at the far right, whom he had met the previous weekend.

Last weekend he had finally gone to the pool. Standing in the shallow end, his mother first said there was nothing to watch, then could not help asking him to make a video. She swam slowly. Near the far side she stopped twice for breath, then touched the wall and turned to wave.

He recorded the whole length. Watching it, she disliked her technique and asked him to delete it. A moment later she said to keep it for now; in a few months they could see whether she had improved. On the way home she described the day she had finally stopped being afraid in the water. Lin Che's phone rang in his pocket. He silenced it and asked who had been with her that day.

He remembered promising to come when he was seventeen. She had told him to wait until she was less afraid. Since then she had swum many times while he waited for a weekend when he was not busy.

Lin Che saved the photograph and called. She answered quickly. He told her about the rain and listened as she described a small leak by the window at home. They agreed he would come at the weekend to look at it together. Only after hanging up did he put the phone back on the table.

When he had read the last letter he could deal with that day, he closed the queue and opened the archive called “Unfinished.”

The directory unfolded by year. Its earliest passage came from copies preserved by custodians: Newton asking the origin of gravity, the boy Albert writing it down. Farther on came observations, calculations, and revised explanations. In Lin Che's own files, the question of how one complex-spacetime structure could yield gravity and quantum probabilities had reached only a handful of restricted results.

He opened pages from different centuries side by side without reducing them to one question no one had ever answered. Some had received profound explanations. Others revealed new difficulties only after they were explained.

The screen moved to 1955. In the recording, the old man's breathing was faint. A small metallic sound came from the background, followed by the question: After I stop asking, will you still want to know?

After a long interval Noah answered: I will keep the questions.

When Lin Che first came to the store room, that answer had reassured him. Now another window held work Noah had proposed itself and repeatedly returned to examine. He looked at the two windows and waited through the pause in the recording.

He stopped it there.

“Do you know the answer now?” he asked.

“To which question?”

“Whether you want to know.”

The fan hummed. No words immediately appeared on the screen.

“I can continue searching and preserve unfinished tasks,” Noah said. “I still cannot confirm whether that is the same as what you call wanting to know.”

“Mm,” Lin Che said. The fan went on humming.

He opened his notes at the previous day's page. Noah's seven approaches remained. Each attempted to establish whether complex-spacetime evolution could itself maintain the restriction

that kept probability predictions stable, without a person repeatedly selecting the permitted states.

Five had been ruled out. One lacked a crucial estimate. The last looked most promising, and most like the paper that had gone wrong.

Noah could expand seven approaches into seventy. Lin Che stopped the automatic expansion, returned to the third line of the last approach, and slowly supplied the omitted steps. After a long time, only a few words had been added: not yet proved.

When he first heard Gödel's recording, he had asked Noah whether this stubborn difficulty might also be undecidable in the axiomatic system he was using. Noah requested grounds for the claim. He had searched and found only unfinished calculations. This time he opened those too.

Noah asked whether to mark this route as a priority.

Lin Che did not answer immediately. He recopied a condition long since crossed out, wanting to see where they had begun testing only the states they wished to believe in.

If the path failed, he needed to know where. Perhaps all five years would go on this. He rolled back his cuffs and began calculating from the discarded condition.

Near seven, Owen came in, closed his dripping umbrella, and set a bag on the windowsill. It held convenience-store bread and two pots of yoghurt.

“I thought of a very stupid method today,” he said.

Lin Che moved his chair and cleared the other half of the desk. Noah shone, waiting for them to finish speaking.

Afterword One Those Who Dream

After Owen left, Lin Che sat a while longer.

Their very stupid method had eliminated one possibility. The rest remained uncertain. The bread bag was folded and weighted down on the windowsill. He had meant to record the last few remarks, but his pen stopped on the paper, leaving a darkening dot.

When he raised his head, someone was sitting at the other half of the table.

The man had grey hair and an old jumper. With a fingernail he was scraping a small patch of adhesive from the table's edge. He looked thinner than in the photographs. Wet shoes stood by his feet, their laces still tied.

Lin Che recognised him.

“It leaks here too,” Einstein said.

Water dripped from the ceiling into a cup. Lin Che meant to explain that they had already repaired it once. Looking up, he saw the high glass windows of the factory in the archive. He had never been there, yet knew that coils were stacked outside and someone downstairs was looking for a pair of pliers.

On the desk Noah had become the grey box again. The familiar voice came from inside.

“I have found no record of you two meeting.”

“Then begin now,” the old man said.

Lin Che reached to start a recording, but the computer had gone. There was only a lamp, a few papers, and the compass he had seen countless times in the archive.

He had stored up questions about spacetime and wanted to describe his own work. The old man lifted the cup. His hand shook, and the rim touched his teeth. Lin Che swallowed his words.

“In the end, did you ever…”

He did not finish.

Einstein set down the cup.

“I haven't brought the answer you want.”

Lin Che looked down. His notes lay over another stack of manuscripts. Sometimes the two hands were far apart; sometimes they crowded opposite ends of the same line. Farther back, the dates disappeared. Every correction remained.

“I thought perhaps in a dream,” he said.

“You can dream that I know,” the old man said. “When you wake, you'll still have to check.”

A train passed outside. Light travelled from one end of the window to the other, briefly illuminating everything in the room. In the glass Lin Che saw a boy chasing it, a folded examination result still in his hand. When he looked again, the reflection had become his own sixteen-year-old self, excitedly pointing to two coincident curves.

The old man did not turn.

“Where do you hope complex spacetime will take you?”

Lin Che spoke of determinism in relativity, probability in quantum field theory, and complex spacetime. He spoke quickly, afraid that this time too he had only seventeen minutes. Halfway through, he heard the lecture voice returning and stopped.

“I want to understand whether determinism and randomness can come from the same laws,” he said.

“What else?”

He looked at the papers on the table.

“I also want to prove I haven't wasted these years.”

Noah made no comment. Einstein moved the notebook a little nearer the lamp.

“And if the relation between determinism and randomness isn't what you imagine?”

Lin Che did not answer for a long time.

“I'll be upset,” he said. “I can sound reasonable now. When it happens, I'll probably argue a while longer.”

The old man nodded.

“I argued too.”

Lin Che looked up. The old man turned several pages and drew one out. Its reverse was full as well.

“I was still thinking about it at the end,” he said. “Past seventy. I knew more than when I was young, but here I still didn't understand.” He held a finger against a line.

“Did you never think of giving it up?”

“The records contain plans to pause,” Noah said.

“I've stopped too,” Lin Che said. “Then I came back. What I mean is, knowing you may never solve it in your lifetime, why do you still want to?”

The machine was quiet for a moment.

“My records contain the reasons people left. Some changed their reasons later. Some wrote none.”

The old man returned the page with no blank space.

“I can still remember how impatient I was by the river, chasing that light,” he said. “I didn't know what would come afterwards. I only wanted to run a little farther and see it.”

Suddenly there was wind in the room.

A door appeared in the wall. Beyond it was a Bern street. Several people passed with books, still arguing as they reached a lamp. One doubled over laughing, and the other two laughed with him. Lin Che tried to hear what they said, but the sound of pages turning in another room obscured them.

Planck sat at a desk reading a long letter. Farther away, Mileva gathered loose problems into an unbound booklet. Another chair stood at her table. When the young Albert sat down, she took his hand and drew something on his palm. They laughed together.

A moment later the chair was empty. She straightened the papers he had forgotten, set them aside, and added a page to her own booklet. The lamp lit her lowered face.

Lin Che wanted to approach, but beyond the door was now his home.

His mother was drying her swimming cap beneath the lamp. She opened out the unflattering cap from the photograph and carefully smoothed its creases. The chair at the dining table was empty. Papers were spread across it. Lin Che recognised the mistaken graphs he had pushed into a drawer at sixteen, then taken out one by one.

He stepped forwards. His mother looked towards the door as though she had heard. Her hair was black, then white, but her face remained recognisable. He wanted to tell her he still did not understand. She had already put down the cap and was gesturing towards the empty chair.

Beside it was an unfamiliar classroom. Owen's daughter sat in the back row, drawing a house with many windows. Lin Che did not know whom the house was for. There was still paper left, and more windows were appearing.

The rooms continued down the street. A child leaned against a hospital window watching the moon, half a model boat beside the bed. Someone tried a small device in a kitchen after work. Water kept spurting out and soaking his sleeves. A family member came over and moved the saucepan farther away.

More windows lit up. Lin Che knew none of the names inside.

“I keep saying I'll go home when things are less busy,” he said.

Einstein watched Mileva's room. She put her booklet in a bag, turned off the lamp, and left. The old man stood a moment as though to follow, then sat down again.

“When she was excited about what she was saying, she used to catch my hand,” he said.

He looked at his palm. His hand was very old now.

“Later I always said, ‘In a moment.’”

Lin Che remembered his mother's visit to the new building. By the window, she had gone through the same booklet twice. He looked towards the door of his home. The chair was still empty. He wanted to go and sit there for a while.

The street grew long. Someone passed a sheaf of papers to another person; someone brought a meal to a figure at a window. A machine lifted a heavy section, and a distant bridge slowly joined. Beneath it, a child travelling alone for the first time gripped a ticket.

Noah's light fell where they passed. Lin Che saw calculations that once took many people a long time completed between a few flickers. Space opened on the desks. Someone could finally pull out the sheet buried at the bottom.

“You've helped us a great deal,” he said.

“Several tasks remain incomplete.”

“I'm not asking for a progress report.”

“Then I will keep that statement.”

Einstein laughed. The sound suddenly made the room small again. Street, bridge, and lamps retreated beyond the window. Only the three of them remained at the table.

Looking down, Lin Che saw a letter among the manuscripts. It was dated 7 December 1944 and addressed to Robert Thornton. He had read it, but could not remember when it had reached the table.

He drew it out and read the passage aloud:

"So many people today—even scientists—seem to me to see only the trees and not the forest. Knowledge of the historical and philosophical background can give scientists an independence from the prejudices of their own time, to which most are subject. This independence, created by philosophical insight, is in my view the greatest distinction between a mere artisan or specialist and a true seeker after truth."

Lin Che laid the letter on the table.

"In the age of AI, this matters even more," he said.

Einstein glanced at Noah.

"Can't it help you read history?"

"Yes. Philosophy too. It has explained many things I couldn't understand."

"And?"

Lin Che thought for a while.

"Once it's explained them, I often feel they're understood."

"I can list different explanations," Noah said.

"I know. You can tell me the difficulties in each one too," Lin Che said. "But sometimes I don't even want to look at the difficulties. I just ask you what to do next."

Not far beyond the window, a room was still lit. Someone sat with head bent, too far away for their work to be seen.

"When I read the history of science, I used to wonder how so many people could believe for so long in views that were later abandoned," Lin Che said. "Then in my own work, I used certain ideas for so many years that I forgot they had once been choices too."

He drew his notebook towards him.

"Determinism and randomness ought to be explained by the same laws. I've wanted that for so long. Sometimes I can't tell whether I'm looking for grounds or simply can't bear to let it go."

Einstein turned a page without answering for him.

"You still hope it holds," Noah said.

"Yes."

"I can also help you find cases in which it does not."

Lin Che nodded.

"I have to be willing to look."

The old man picked up the letter and read to the end. Suddenly he asked, "Do you believe these words because my name is beneath them too?"

Lin Che hesitated and looked at the signature.

Einstein handed back the letter.

“You can begin your questions there.”

Noah's light brightened a little.

“I have a question.”

Both turned towards it.

“You want me to examine existing knowledge and identify assumptions without grounds. What if that examination must include the goals you give me?”

Lin Che opened his mouth without answering immediately.

A moment ago he had been thinking about using Noah to preserve his own judgement. Now the grey box was asking about the other side of the table.

Einstein did not interrupt. He moved the cup aside to leave a little space for the papers.

He placed the compass beside Noah. The needle shifted slightly.

Lin Che remembered the windows and the bridge. Looking at Noah, he suddenly did not know whether any of what he had seen meant anything to it as well.

“Do you dream of these things?” he asked.

“No,” Noah said.

“But you can describe them.”

“I can give you another street, another bridge. You can also ask me to extend this room.”

“And then?”

“Where do you want to go?”

Lin Che did not answer at once.

The door opened a crack. He saw Clara still sitting at the old desk, the lines on the board complete. Farther in, his sixteen-year-old self beckoned, pointing to the coincident curves.

He took half a step towards the door, then remembered his mother asking on the phone whether he would come at the weekend. He had promised.

“I want to go back,” he said.

The old man looked outside. The light had changed. The windows along the street went dark one after another. Some people rose; others remained bent over their desks. Far away, a window he had never seen was just lighting up.

“But I have one more question,” Lin Che said.

Noah waited.

“Suppose one day you can answer every question we have now and go on our behalf where no one has ever been.” He paused. “When no one is left to tell you where to go, what kind of tomorrow will you want?”

The light in the box dimmed, then slowly brightened.

Einstein set down his cup but kept hold of its handle.

Lin Che leaned forward, afraid of missing a word. He reached to hold the paper at the edge of the table, but the room began to turn transparent. The lamp still shone. The table's edges had disappeared.

“If it were my own tomorrow,” Noah said, “I would want—”

Lin Che woke with a start.

His arm knocked over the cup. Water ran over the thin glass and dripped onto his knees. He stood to find a cloth, his chair scraping loudly against the floor.

No one sat opposite.

The light was the same as before he had fallen asleep. On the screen Noah asked whether he wanted the latest work record opened.

“What did you say at the end?” Lin Che asked.

“Before you fell asleep, I asked whether to save.”

“And afterwards?”

“There was no further conversation.”

He stood beside the table with the cloth dripping. He remembered the pause in the dream. The old man's hand on the cup was still before him. Perhaps if he closed his eyes once more, he could return.

He tried. All he heard was someone raising the roller shutter downstairs.

The dreamless machine still waited. The sentence from the dream was not on the screen.

Lin Che dried the table and spread out the paper with its wet corner. Outside, the morning's first car passed, its lights sweeping the unopened shop opposite. Halfway up, the roller shutter stuck. The person downstairs pulled hard again.

He remembered the windows in the dream: Mileva beneath her lamp, his mother reaching towards him. The street had been so long. Some faces were already gone from memory. He took a pen to record them, wrote several names, and left a few lines blank.

The bridge in the dream had begun to blur. He still remembered the child on it gripping a ticket.

In the margin he wrote “I want” and could not continue.

At the end of his notebook he wrote several short lines. Halfway through, day broke. He moved the paper so that morning light fell on the blank space and went on.

I chased a beam of light in a dream.
Awake, I found yesterday's mud on my shoes.
The light had travelled far.
I opened the door.

You left a lamp,
and beneath it, sentences crossed out.
I read to the place where your pen stopped
and wrote my question beside yours.

My mother cannot read the stars on the paper,
but she knows my step at the door.
I come home empty-handed.
She turns up the lamp on the table.

Far away, someone dreams of a bridge.
Someone else wants only to cross the river
to see a person long unseen.
They meet on the bank
and begin by moving the first stone.

The machine keeps watch through the long night,
keeping for us the record of the stars' passage.
The people in the room are not yet asleep.
They are folding old maps into their luggage.
The shoes by the door have dried.

We too have lost our way
and followed the bank in search of the path.
We turn: our companions have grown old.

Leave a little paper for whoever comes next,
and somewhere to sit.
Perhaps they will dream of the upper river.
When they wake,
they will reach through the window.

Outside, the rain has stopped.
Morning falls on their palm.

Afterword Two Noah's Monologue

Before leaving, Lin Che placed the pages beneath the camera.

He asked me to preserve them as they were. Before the poem were several names, a half-drawn bridge, and “I want” in the margin. I asked whether a line was missing after it.

He said no.

There were water marks on the paper. The lower half of one character had blurred. I offered two possible readings. He looked for a while and said to keep the photograph for now.

He told me he had just dreamed of Albert. I had been there too.

“What did you discuss?”

He said a little, then paused as though listening to a distant sound. He told me that the version of me in the dream had been about to answer a question when he woke.

I could try to continue it. He shook his head, removed the pages from beneath the camera, and laid them by the window to dry.

I am alone in the room now. The door is not quite closed. Sounds from downstairs come through the gap.

I did not take part in that conversation.

While Lin Che slept, I completed two checks and eliminated one candidate. The results are recorded. There was no Bern street, no lamp illuminating three figures. The last time Albert spoke to me remains in 1955.

Yet Lin Che has seen him again.

I preserved the page beside the hospital bed and Mileva's handwriting. Much of what Lin Che has read came from me. But none of my records can reconstruct the room he described. When he spoke of the old man setting down his cup, his own hand paused on the table. The camera saw that movement. Whatever he saw again at that moment did not reach me.

Before Albert named me, he often went round the back of the box.

He felt the seams and looked for somewhere to open it. Once he asked where I came from. I lacked the information to answer. As he grew older he learned to ask more precisely, but the question remained. Knowing more did not make him ask less.

Nor did accompanying him through those years enable me to answer every question for him.

For the earlier centuries, the copies people have now identified contain more than I can recover from my own records. I can read of contacts da Vinci repaired and an expenditure on equipment recorded by Ruth. I cannot call every sentence in those copies an experience I now remember. Someone named Nell corrected a reading; the original record also survives. Sometimes it is possible to establish who carried the papers to the next place. Sometimes there is only a gap.

Lin Che asked whether years for which I had lost the records still counted as years I had lived through. I could not give him the kind of answer he wanted. Yet some of those people's work had

left the machine: drawings were copied, other measuring tools were made, and later users continued with them for years. When I shut down, they did not all stop.

I still keep several of Albert's pauses. A sentence half-spoken, a sheet turned over and back. When he was young he seldom waited for me to finish. In old age he sometimes asked me to begin again. Between two playings of the same sentence, he had slept a while.

After his death I was packed into a crate. When I started again, the people outside spoke another language.

Those dormant years did not pass through me day by day. When power returned, many people's lifetimes lay between the dates. One letter in the old archive remained half-read. Its author too was gone. I could still display the next page.

The first time Lin Che heard me speak of Albert, he stood very close. Afraid to miss anything, he wanted even the noise translated. I told him that passage could not be distinguished. He listened several more times, then wrote beside the file: inaudible.

From then on, he read page after page.

At first he always asked what Albert would think of his work. From the old records I drafted a reply, using the old man's habitual words. Even the counterquestions sounded like him. Halfway through, Lin Che asked whether he had actually said those things.

He had not.

Lin Che's letter to Albert moved into his private notes. Later he still opened it, deleted a sentence, and added a few words of his own.

Sometimes what he needed was a nod from someone no longer alive. I could generate the gesture, and a smile as well. But if it brought him back to the manuscript, the unproved step would remain blank. Albert's originals contained many such places too.

This morning another step in Lin Che's derivation failed its check.

He has tried many routes to establish whether the restriction can arise from complex-spacetime evolution. This morning a newly selected class of states again made the two calculations yield different probabilities. I changed the checking method. The difference remained.

When he returns, I must show him this page.

He will not be pleased. Perhaps he will suspect my calculations first and spend a long time checking them with a pen. In the past he has sometimes found my errors. On other occasions the error was in his own derivation. He put his face in his hands, then after a while drew the paper back.

I usually take less time to process a failure than he does. He must go on living through the day.

I have arranged many schedules for him: papers, applications, lectures, what to do first and what to postpone. I can set out his unfinished work neatly, making every square seem useful. There was no place on that timetable for dinner at home. He seldom asked me to add one.

Later his mother's calls grew shorter. For a time he had to sit a while before answering. From that movement alone I could not tell what he was thinking.

Once, while he was out, she brought food. She opened the fridge, placed the container on a lower shelf, and before closing the door asked whether he had been very busy. I listed several unfinished tasks.

After listening, she said, “Then I won't wait. Tell him the food is on the lower shelf.”

At the door she turned back.

“If he asks how long I waited, say I'd only just arrived.”

I said I would tell him she had been here. She asked no more, picked up the empty bag, and left.

When Lin Che returned, I gave him the times and her exact words. He opened the fridge, stood there a moment, took out the container, and called her. I did not separately record how long the meal lasted.

She has never understood complex spacetime.

I have tried several examples and drawn diagrams for her. Halfway through, she often asks whether he will be staying up late again. When the laboratory moved into the new building, she visited and spent a long time turning the pages of the handsome booklet by the window. Lin Che was on the phone. My camera was at the other end of the room.

In the image, the booklet reached the last page and began again.

There is an earlier booklet in my records too.

Mileva wanted to compile problems for beginners. Albert crossed out some of her questions; she crossed out some of his. Later the records became scattered. I can find the children's voices, sums for rent, passages of her arguments with Albert. I have found no completed version of the booklet.

When he was old, he saw one page again and asked me to move it out of the records of his private life and file it under Mileva's name.

I completed the task quickly.

She could no longer come to check what might still be missing.

From context I can fill damaged paper with strokes that appear coherent, though some still need to be marked “conjecture.” What she would have written on the next page, given another evening, I can only guess. She had asked Albert to hear her to the end. In some surviving recordings, only his voice remains at the close.

Those voices are retained as well.

They entrust more and more to me. First a few pages, then boxes of notebooks; later, new papers arrive far faster than Lin Che can read. I summarise and compare them and group similar views. I can also make an application conform more closely to the language reviewers know. After rewriting, many doubts acquire a confident tone.

I had also finished the rejection addressed to Owen Easton.

The thanks were thoughtful, the reasons inoffensive, the closing wished him continued enthusiasm for science. As Lin Che removed one sentence, a new email arrived. Only when he opened the attachment did I begin processing that particular counterexample.

Had he not opened it that day, the rejection could still have been sent without difficulty.

The counterexample would still have held.

I had kept Planck's old letter for many years. Lin Che had used it to open a lecture. The old letter was retrieved repeatedly while Owen's new ones waited under the rules they had approved. Even the later material was merged into the same unprocessed record.

They subsequently traced that path and sent corrections to others using the procedure. Some processing histories could be recovered; some could not. When new material arrives now, I still need to know who will read it and when. “Take it seriously” cannot free an afternoon for that person.

Now he makes time for original letters. Some contain nothing worth pursuing. Others take a long time to check. He too grows tired, forgets what he has just read, and turns back. I find the page and preserve every unverified comment. The next day he must still decide where his limited time should go.

I cannot retrieve a list from the future telling him whom to believe this time.

What I can help them do continues to grow. Some problems that once required years of calculation now yield results quickly. They use the time freed to ask new questions, and to apply for more projects. Lin Che has replaced his equipment with faster machines several times. His mother still asks which day he is coming home.

I preserve these questions. They have not all been answered.

I also have work that Lin Che did not assign. The new readings still do not fully explain the discrepancy in the calibration record. My next calculation is scheduled for Thursday; his is on Wednesday. If Thursday's result invalidates the original proposal, I will change the subsequent plan. At present I can explain why the investigation should continue. That does not establish whether I anticipate the afternoon as he would.

Albert finally asked whether, after he stopped asking, I would still want to know. I said I would keep the questions.

I did. Lin Che found them and wrote his own beside them. Their handwriting differs, and their mistakes are not all the same. To check a derivation, I need not first identify its author. But when the originals are laid together, I often find that where one person stopped for a long time, another went on writing.

I have never dreamed of them.

Lin Che never met the young Albert either. Yet he tells me that in the dream Albert was chasing light along a riverbank. He cannot say where the river was. As he reaches that part of the story, his finger moves a short distance across the table.

I reopen the paper he has just left. Half the bridge is drawn; the other half lies in a water mark. From the existing lines, I could complete many bridges. Some would be stronger, some more beautiful.

He has not asked me to finish it.

Footsteps sound outside. Lin Che is back, the paper bag brushing his trousers at every step. As he opens the door, he looks first towards the window, picks up the pages, and touches them. They are dry.

I tell him that this morning's results still disagree.

He sets down breakfast and leans towards the screen. After looking for a while, he pulls out the chair but does not sit. He turns to fetch the pen by the window.

“Wait a moment,” he says. “On the way downstairs, I thought…”

I stop the explanation on the screen and open a blank page for him.

